From Spatial AI to Physical AI: Learning to Represent, Reason, and Act in 3D

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Photogrammetry and Remote Sensing (PRS)
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Outline

- **About me**

- **Introduction**

- **From Spatial AI to Physical AI: Learning to Represent, Reason, and Act in 3D**
  - C3G: Learning Compact 3D Representations with 2K Gaussians
    - CVPR 2026 # Compact 3D Gaussian Splatting # 3D Reconstruction # 3D Scene Understanding
  - Imagine3D-LLM: Teaching MLLMs to Imagine 3D Scenes Before Answering
    - NeurIPS 2026 # Compact 3D tokens # Mental Reconstruction # Spatial Reasoning
  - Geometric Action Model for Visuomotor Control
    - Arxiv 2026 (under review) # WAM # VLA # Manipulation # Geometric Foudnation Model # World Model

- **Conclusion**

- **Future Works**

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About Me

•Sunghwan Hong (홍성환)

- M.S/Ph.D integrated student advised by Paul Hongsuk Seo and Seungryong Kim
- Postdoc researcher @ ETH AI Center, CVG, PRS



• Education

- 2021.03-current Integrated M.S./Ph.D. in CSE, Korea University
- 2015.03. 2021.02 B.S. in CSE, Korea University

•Professional Experience

- 2025.07-current Postdoc @ ETH Zurich
- 2025.03. - 2025.06 Postdoc @ KAIST
- 2023.04. - 2023.10 Intern @ Microsoft Research Asia

•Research Interests

- Robotics/Egocentric Vision
- VLMs, VLAs, VLNs,
- Visual Correspondence
- 3D/4D Reconstruction and Understanding
- Embodied AI

•Awards

- **Google Ph.D. Fellowship** (2023), Machine Perception, Speech Technology, and Computer Vision
- **ETH AI Center Postdoc Fellowship** (2025), **KAIST Postdoctoral Fellowship** (2025), **Post-Doc Domestic Overseas Training** (2025)

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Introduction

What is Image Matching?



Source Image



Target Image

Why is Image Matching Useful?

SfM and MVS



Sparse model of central Rome using 21K photos produced by COLMAP's SfM pipeline.



Dense models of several landmarks produced by COLMAP's MVS pipeline.

[CVPR Tutorial by Balntas]

Schonberger et al., "Structure-from-Motion Revisited," CVPR, 2016

Why is Image Matching Useful?

SLAM

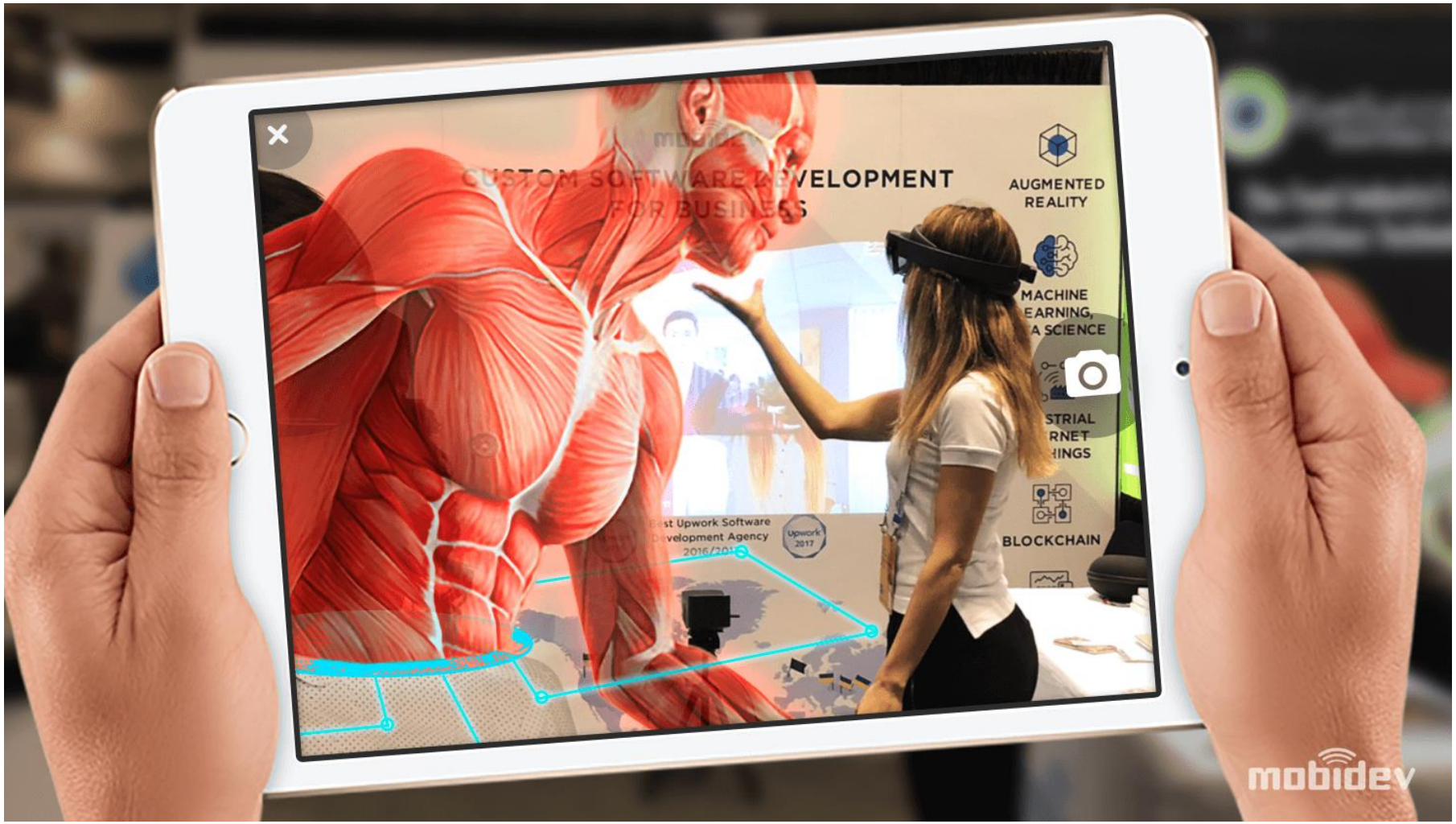


[CVPR Tutorial by Balntas]

Mur-Artal et al., "ORB-SLAM2: an Open-Source SLAM System for Monocular, Stereo and RGB-D Cameras," Arxiv, 2016

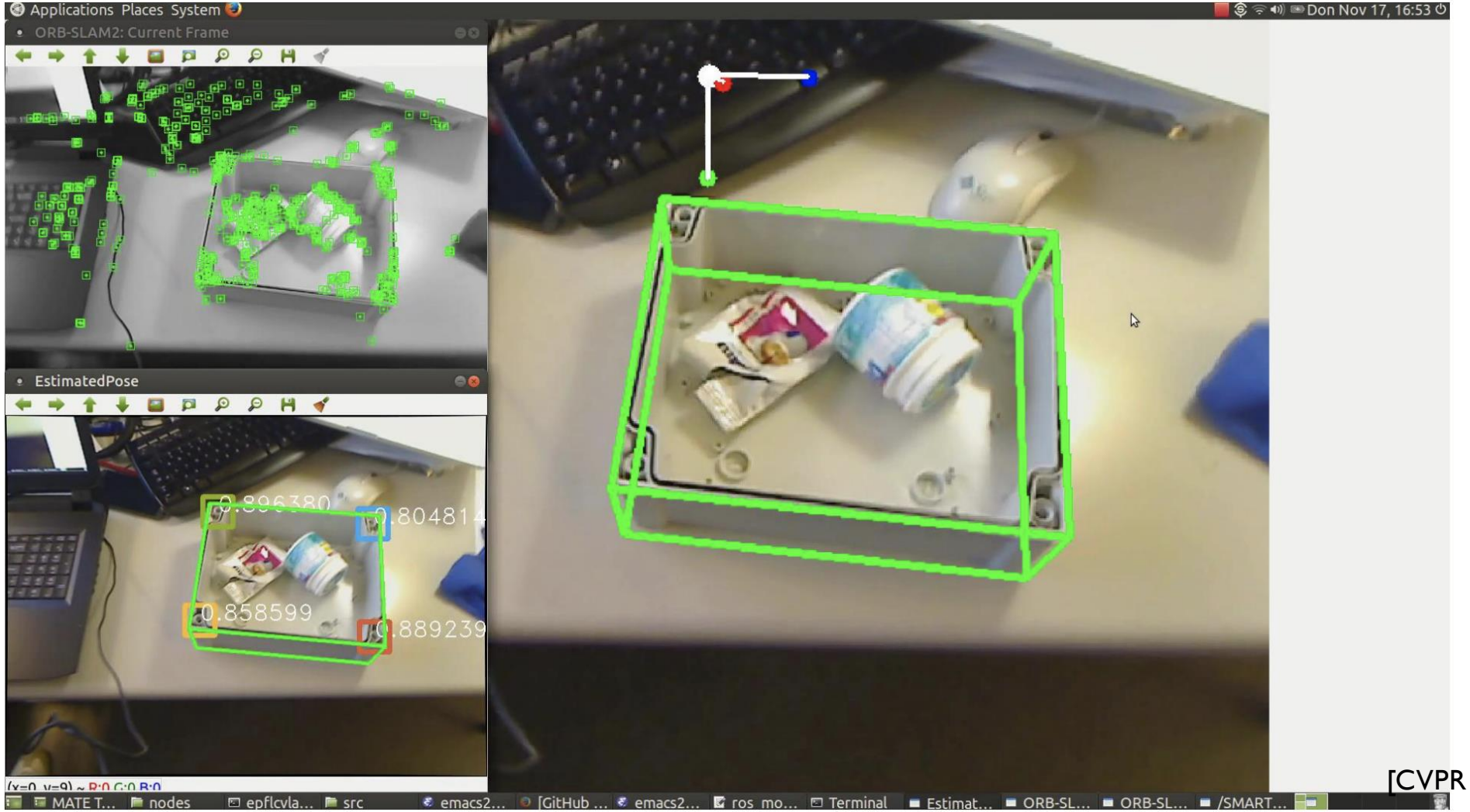
Why is Image Matching Useful?

Augmented Reality



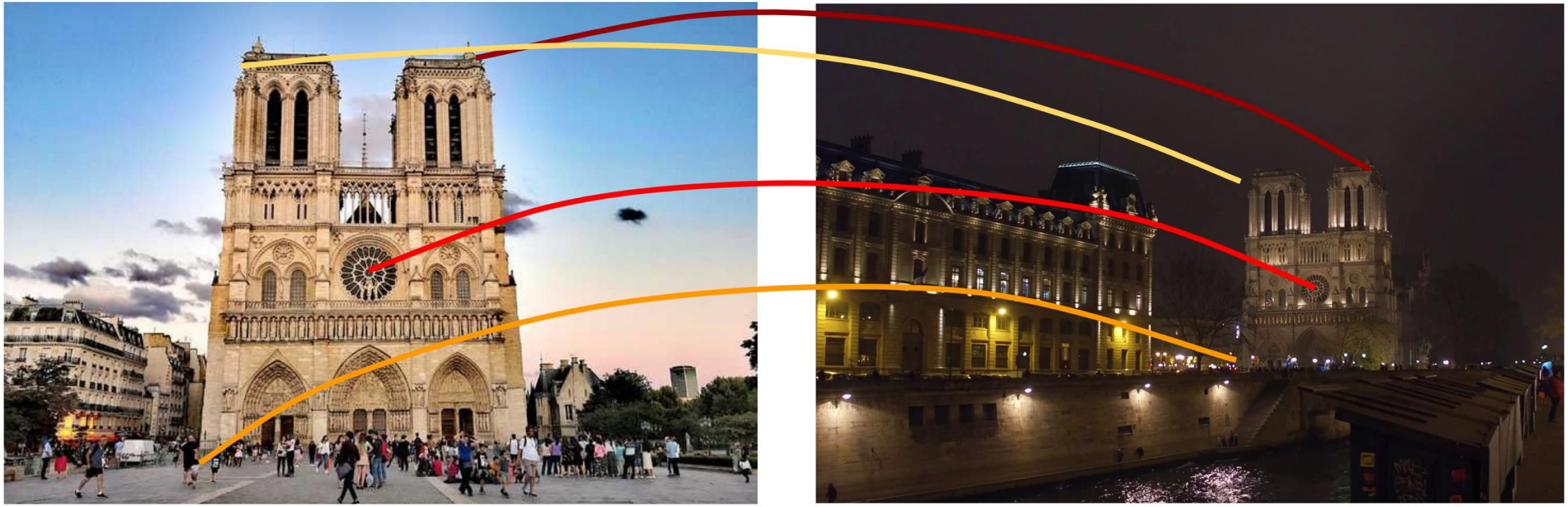
Why is Image Matching Useful?

Tracking Objects



[CVPR Tutorial by Balntas]

Challenges in Image Matching



day-night matching

image credit by Rocco

Challenges in Image Matching



changes across time

image credit by Rocco

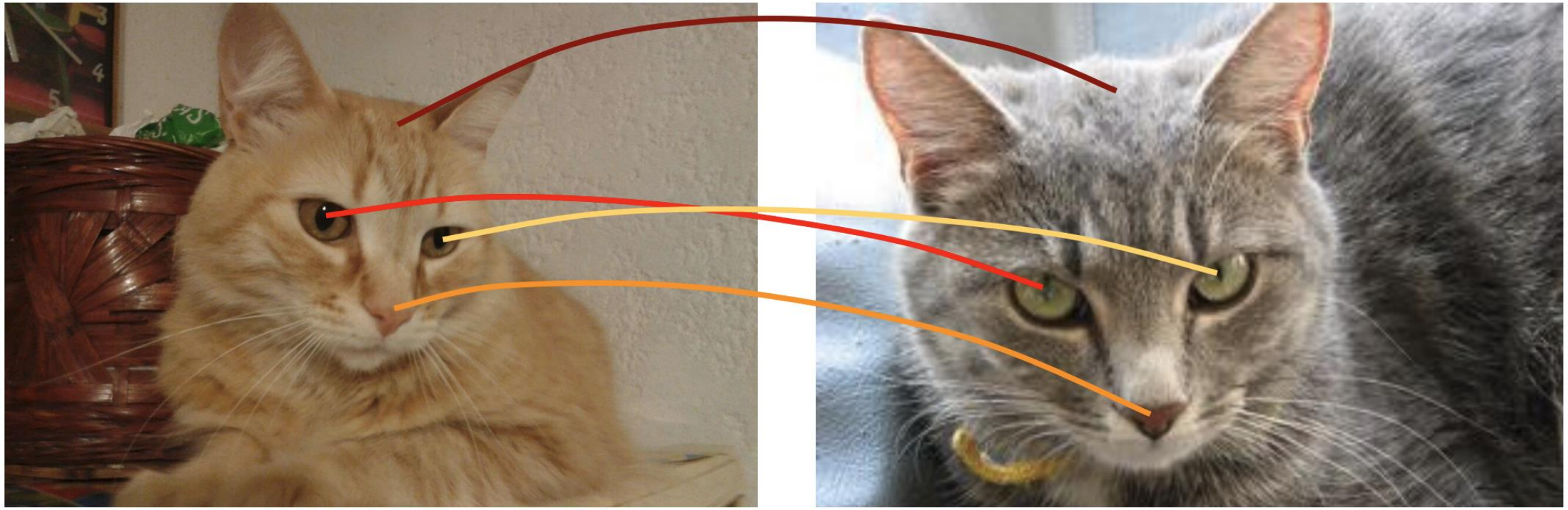
Challenges in Image Matching



repetitive structures

image credit by Rocco

Challenges in Image Matching

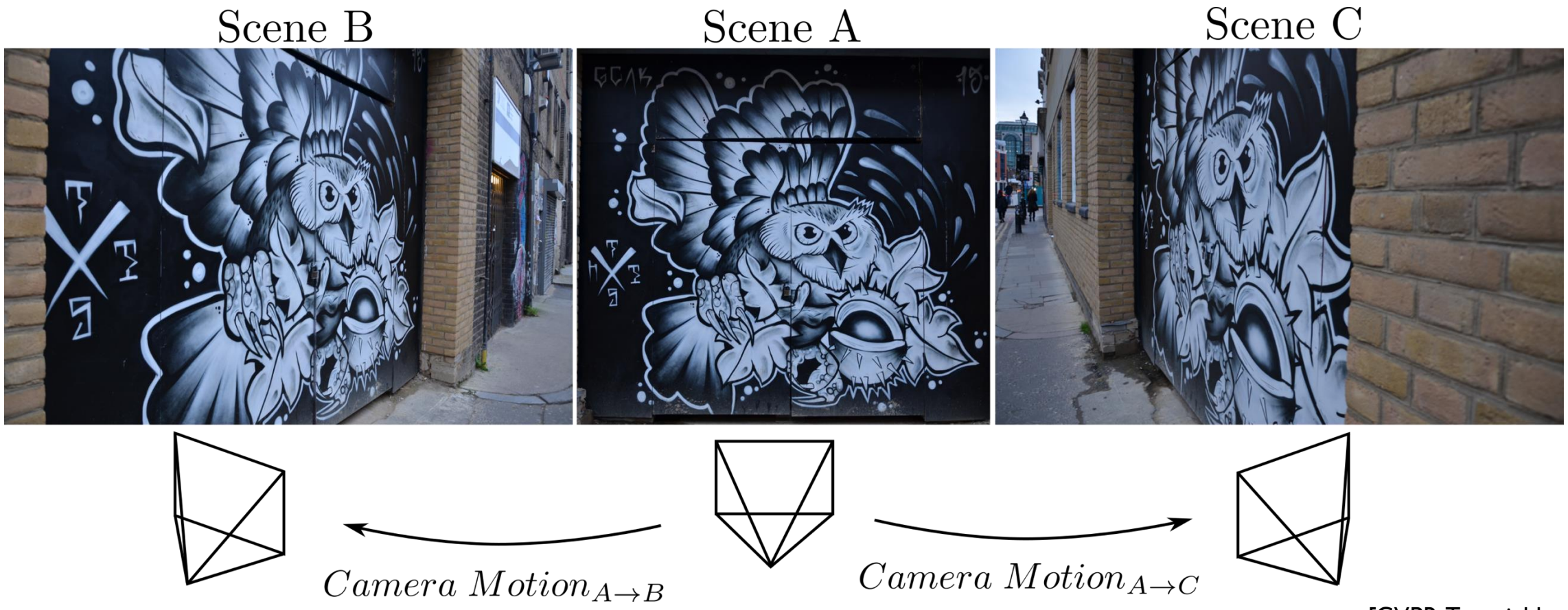


intra-class variation

image credit by Rocco

Image Matching- Practicality

Matching a set of images enables us to ***"recover"*** the geometry of the world from individual images.



[CVPR Tutorial by Balntas]

Hartley and Zisserman, "Multiple View Geometry in Computer Vision," 2000

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C3G: Learning Compact 3D Representations with 2K Gaussians

Honggyu An*, Jaewoo Jung*, Mungyeom Kim, Chaehyun Kim, Minkyeong Jeon, Jisang Han, Kazumi Fukuda, Takuya Narihira, Hyuna Ko, Junsu Kim, Sunghwan Hong †, Yuki Mitsufuji †, Seungryong Kim†

- CVPR'26

Contents

I Problem Definition

II Approach

III Solution

Problem Definition

Problem Definition

Given ***sparse unposed multi-views*** as input, we want to estimate ***3DGS*** that compactly r epresents the scene for ***novel view synthesis*** and ***3D scene understanding***.

So three things to understand the task…

1. Input: sparse and unposed multi-view images
2. Representation: 3D Gaussian Splatting
3. Task : novel view synthesis and 3D scene understanding

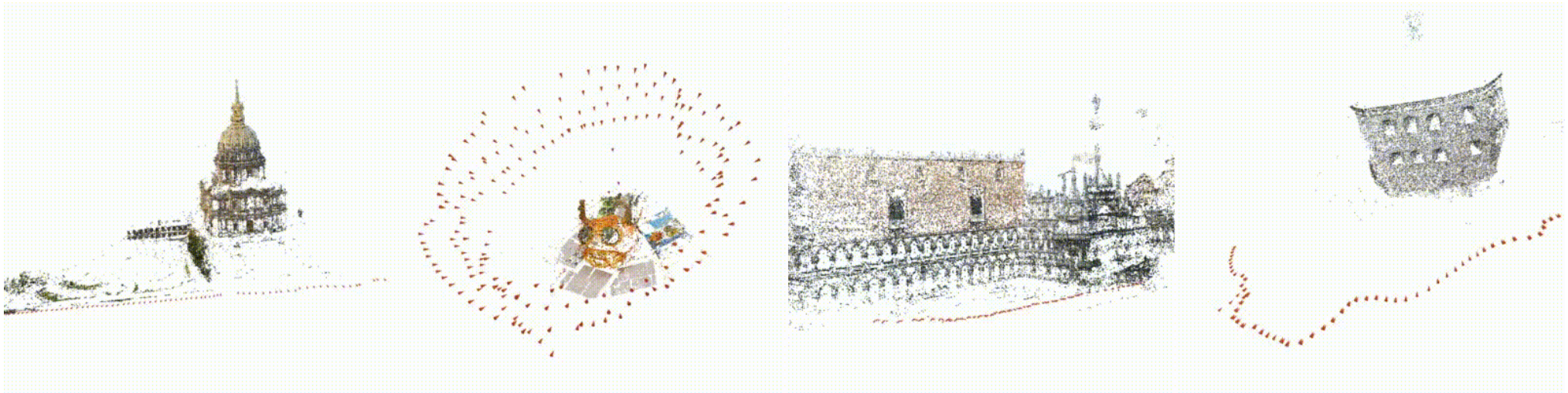
Problem Definition - Input

Given ***sparse unposed multi-views*** as input, we want to estimate ***3DGS*** that compactly r epresents the scene for ***novel view synthesis*** and ***3D scene understanding***.

Q. What do we need to reconstruct 3D scenes?

-> ***Dense*** number of multiple ***posed*** images.

-> Requires slow per-scene optimization to obtain camera poses.



Schonberger, Johannes L., and Jan-Michael Frahm. "Structure-from-motion revisited." Proceedings of the IEEE conference on computer vision and pattern recognition. 2016.

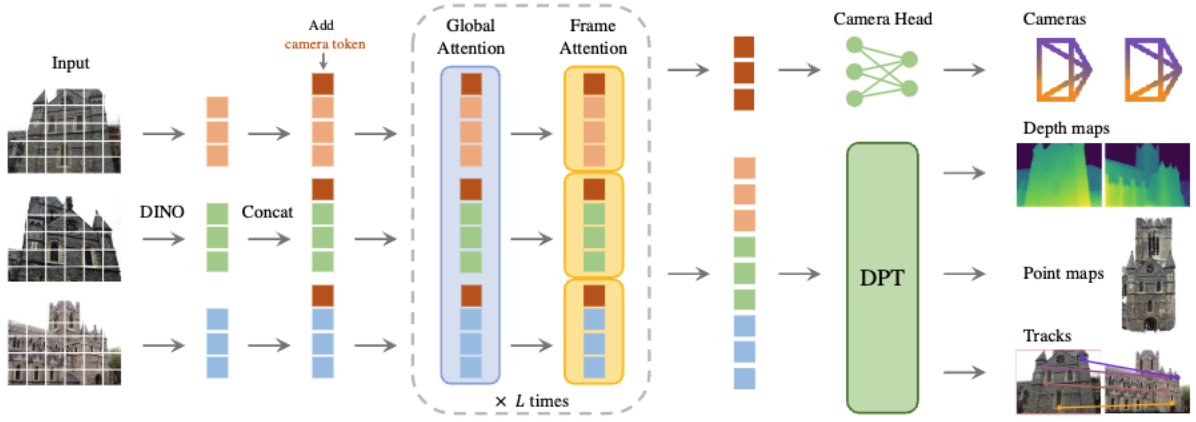
Problem Definition - Input

Given ***sparse unposed multi-views*** as input, we want to estimate ***3DGS*** that compactly r epresents the scene for ***novel view synthesis*** and ***3D scene understanding***.

Q. How can we get rid of the constraint of dense images?

-> Build a model with ***priors*** learned from ***large-scale multi-view data***!

-> Get rid of slow per-scene optimization, and requirement of dense views



Architecture



Feed-Forward 3D Estimation

Wang, Jianyuan, et al. "Vggt: Visual geometry grounded transformer." Proceedings of the Computer Vision and Pattern Recognition Conference. 2025. (Best Paper Award)

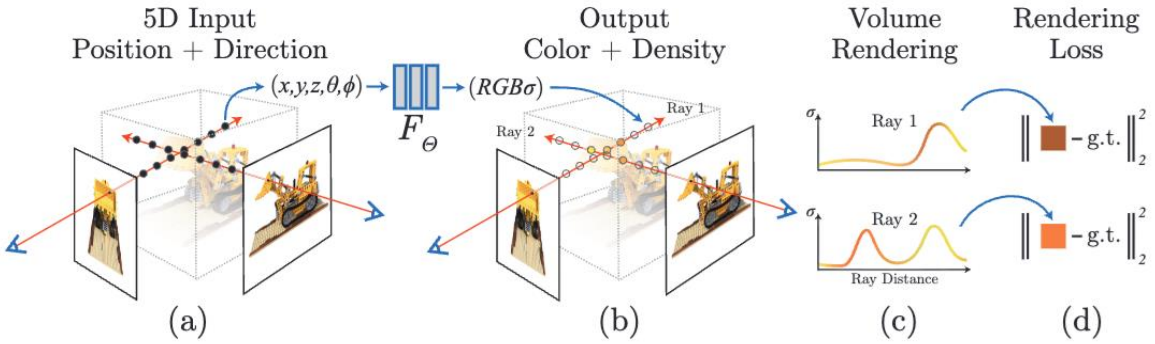
Problem Definition - Representation

- Given ***sparse unposed multi-views*** as input, we want to estimate ***3DGS*** that compactly represents the scene for ***novel view synthesis*** and ***3D scene understanding***.

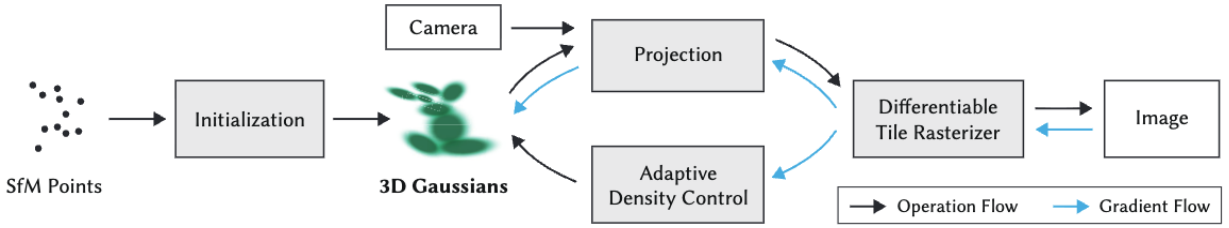
Q. How can we represent the 3D scene to solve various downstream tasks?

-> Neural Radiance Fields, 3D Gaussian Splatting

-> Represent the scene with a neural network (MLP), or 3D Gaussian blobs



Neural Radiance Fields



3D Gaussian Splatting

Mildenhall, Ben, et al. "Nerf: Representing scenes as neural radiance fields for view synthesis." Communications of the ACM 65.1 (2021): 99-106.
Kerbl, Bernhard, et al. "3D Gaussian splatting for real-time radiance field rendering." ACM Trans. Graph. 42.4 (2023): 139-1.

Problem Definition – Task

- Given ***sparse unposed multi-views*** as input, we want to estimate ***3DGS*** that compactly represents the scene for ***novel view synthesis*** and ***3D scene understanding***.

Q. What is novel view synthesis?

-> Given a set of images, we want to synthesize an image as it was seen from a novel viewpoint!



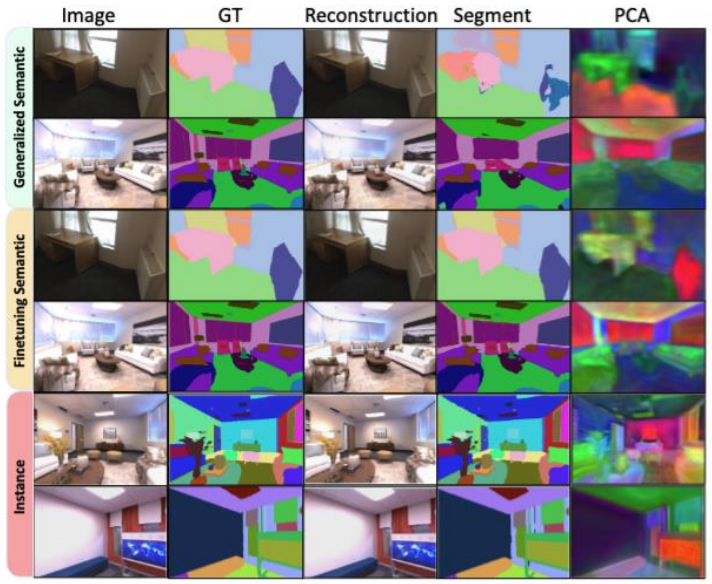
Choi, Inchang, et al. "Extreme view synthesis." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2019.

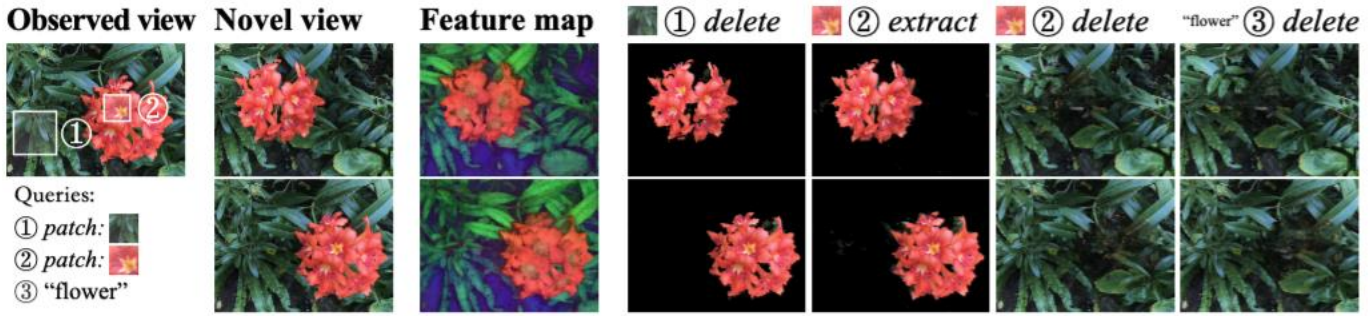
Problem Definition – Task

- Given ***sparse unposed multi-views*** as input, we want to estimate ***3DGS*** that compactly represents the scene for ***novel view synthesis*** and ***3D scene understanding***.

Q. What is 3D scene understanding?

-> Given a set of images, we want to understand (segment) the images in the scene and enable novel-vie w segmentations!

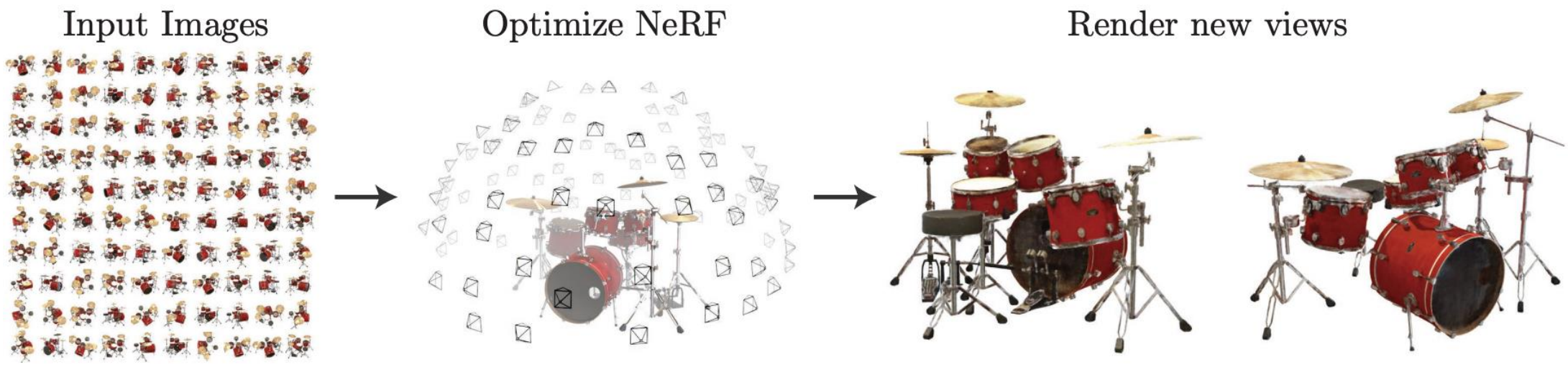




Kobayashi, Sosuke, Eiichi Matsumoto, and Vincent Sitzmann. "Decomposing nerf for editing via feature field distillation." Advances in neural information processing systems 35 (2022): 23311-23330.

Problem Definition

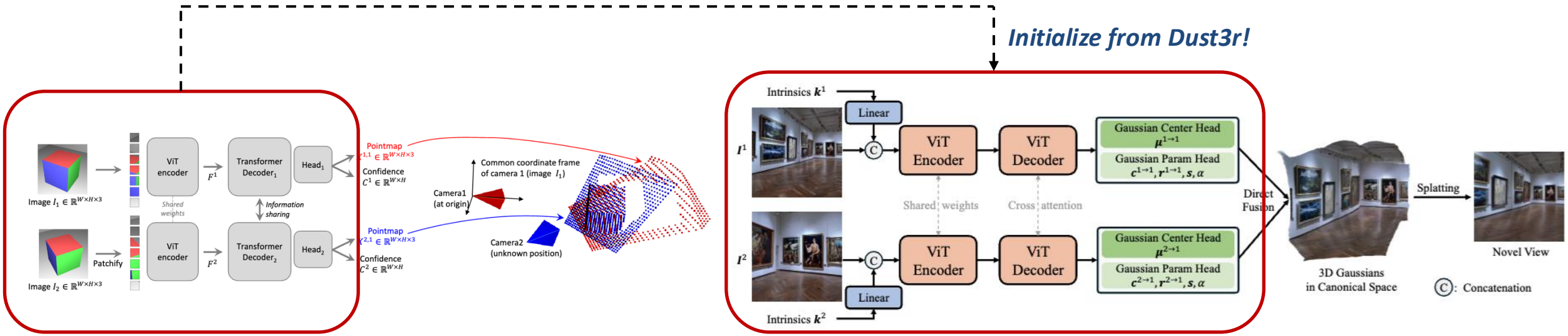
- Given ***sparse unposed multi-views*** as input, we want to estimate ***3DGS*** that compactly represents the scene for ***novel view synthesis*** and ***3D scene understanding***.
- Q. To train 3DGS or NeRF, we typically need dense multi-views.
- -> This also requires dense multi-view images & long per-scene optimization



Mildenhall, Ben, et al. "Nerf: Representing scenes as neural radiance fields for view synthesis." Communications of the ACM 65.1 (2021): 99-106.

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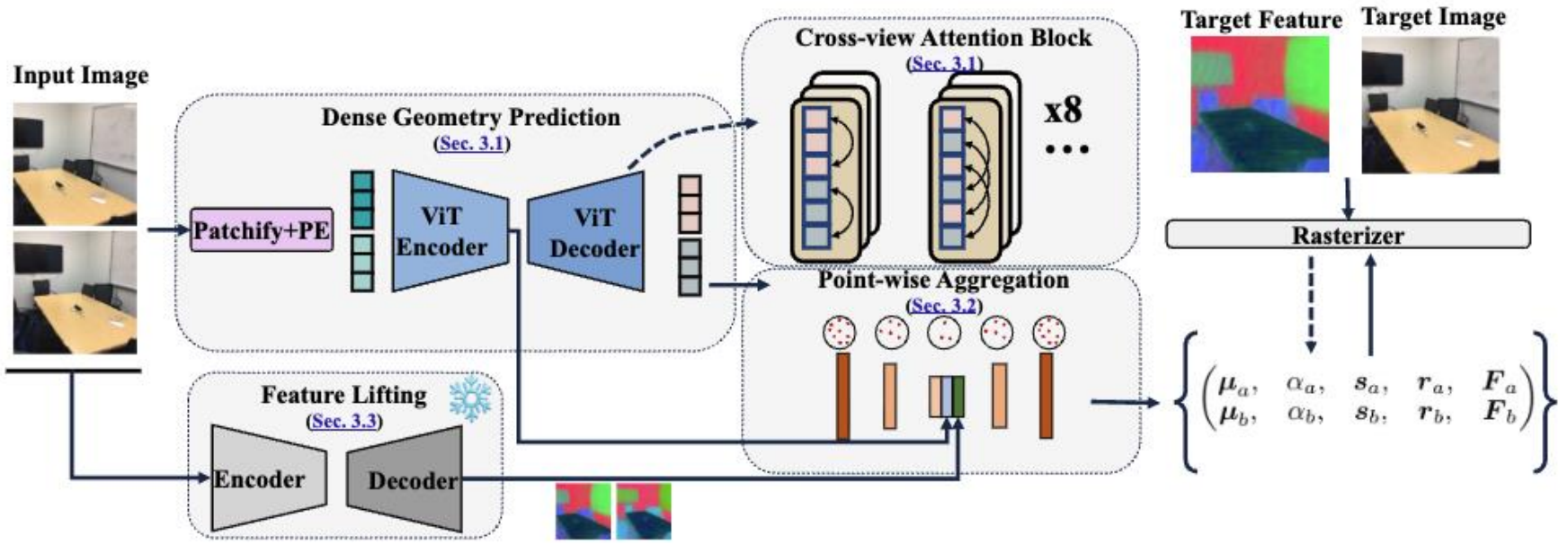
- Given ***sparse unposed multi-views*** as input, we want to estimate ***3DGS*** that compactly represents the scene for ***novel view synthesis*** and ***3D scene understanding***.
- Q. To train 3DGS or NeRF, we typically need dense multi-views.
- -> This also requires dense multi-view images & long per-scene optimization
- -> Recent works have proposed using large-scale foundation models as backbones



Wang, Shuzhe, et al. "Dust3r: Geometric 3d vision made easy." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024.
Ye, Botao, et al. "No pose, no problem: Surprisingly simple 3d gaussian splats from sparse unposed images." arXiv preprint arXiv:2410.24207 (2024).

Baseline – Large Spatial Model

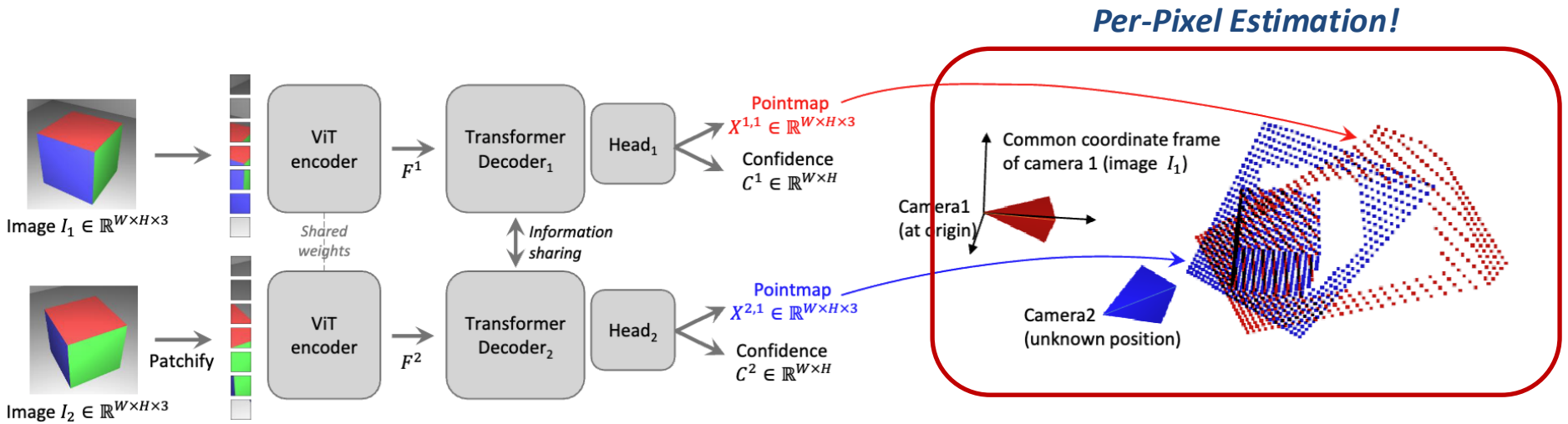
- Given ***sparse unposed multi-views*** as input, we want to estimate ***3DGS*** that compactly represents the scene for ***novel view synthesis*** and ***3D scene understanding***.
- From the learned 3D Gaussians, lift semantic features for 3D scene understanding



Fan, Zhiwen, et al. "Large spatial model: End-to-end unposed images to semantic 3d." Advances in neural information processing systems 37 (2024): 40212-40229.

Baseline – Large Spatial Model

- What are the problems of existing models?
- -> They estimated per-pixel Gaussians, which lead to degraded novel view synthesis per formance and degraded 3D scene understanding.

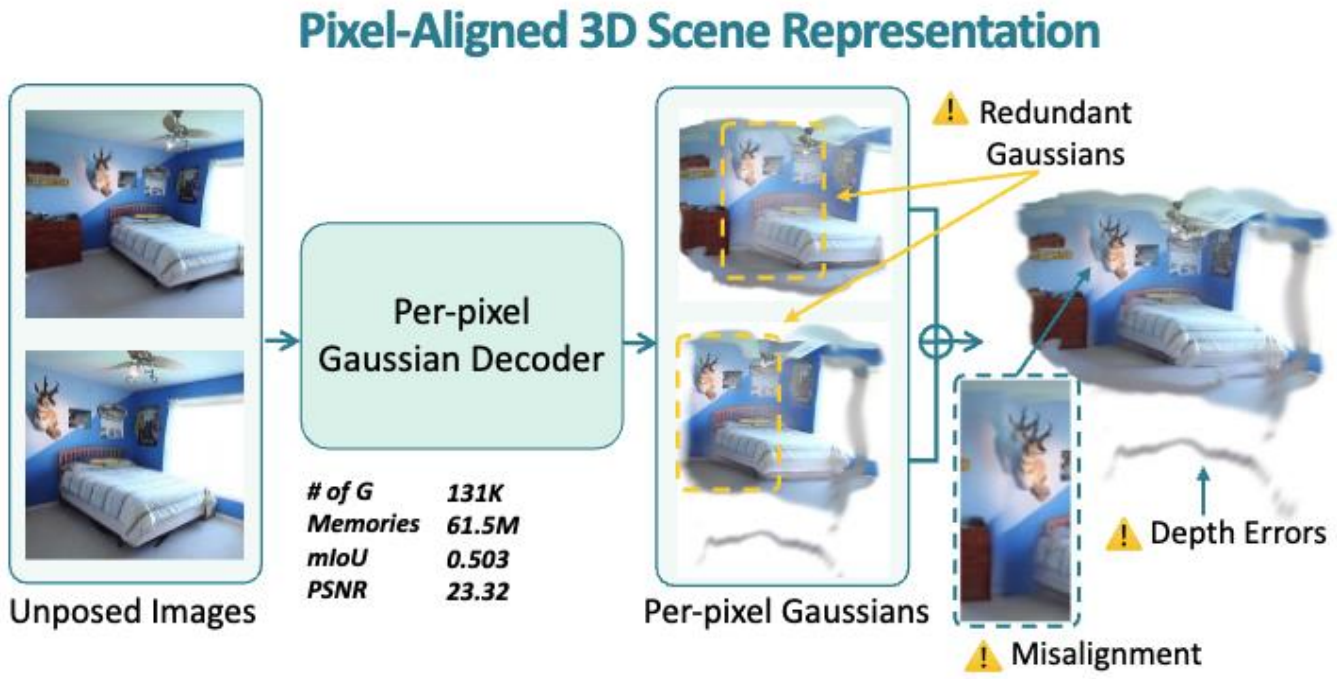


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Baseline – Large Spatial Model

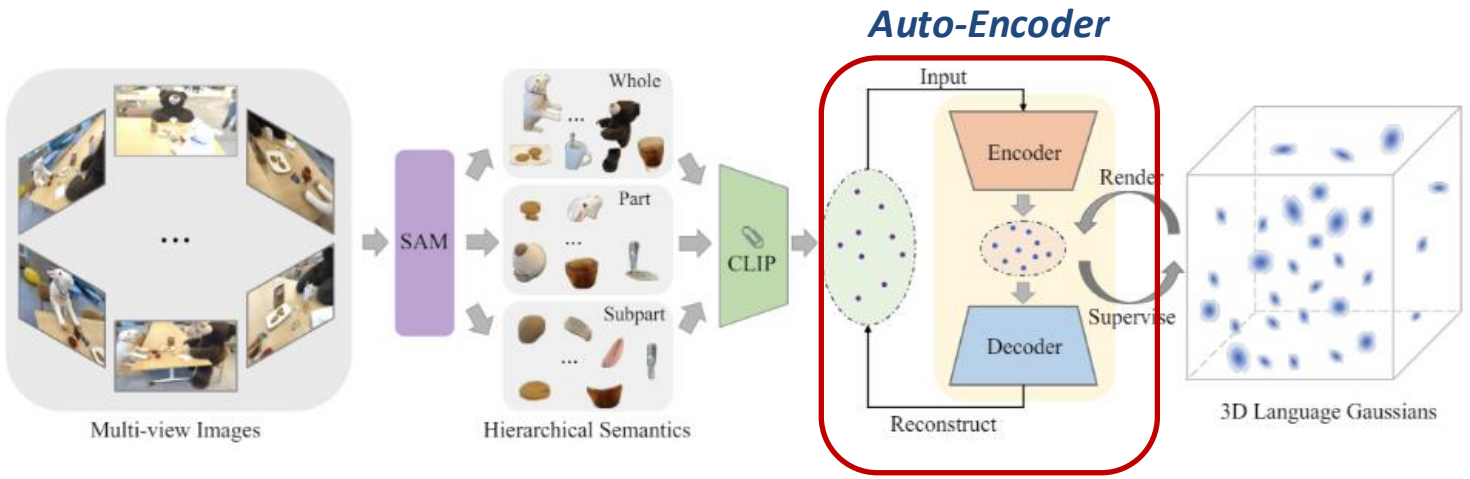
•What are the problems of existing models?

•-> Per-pixel Gaussians often lead to mis-aligned Gaussians, causing artifacts and noisy i mages when rendering to novel views.



Baseline – Large Spatial Model

- What are the problems of existing models?
- -> Per-pixel Gaussians results with too many Gaussians, which results with large computational overhead. This requires an auxiliary model to compress the feature dimensions.
- -> This leads to less computation, but also results in information loss.

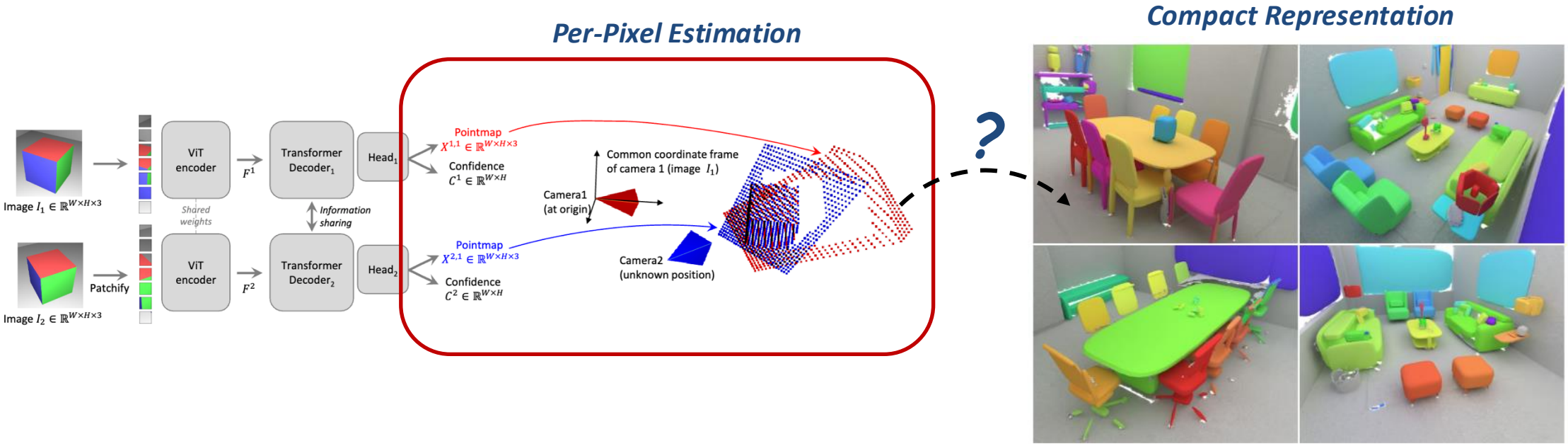


Qin, Minghan, et al. "Langsplat: 3d language gaussian splatting." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024.

Problem Definition

Given sparse unposed multi-views as input, we want to estimate 3DGS that ***compactly*** r epresents the scene for novel view synthesis and 3D scene understanding.

Q. Do we really need such per-pixel, dense scene representations?



Approach

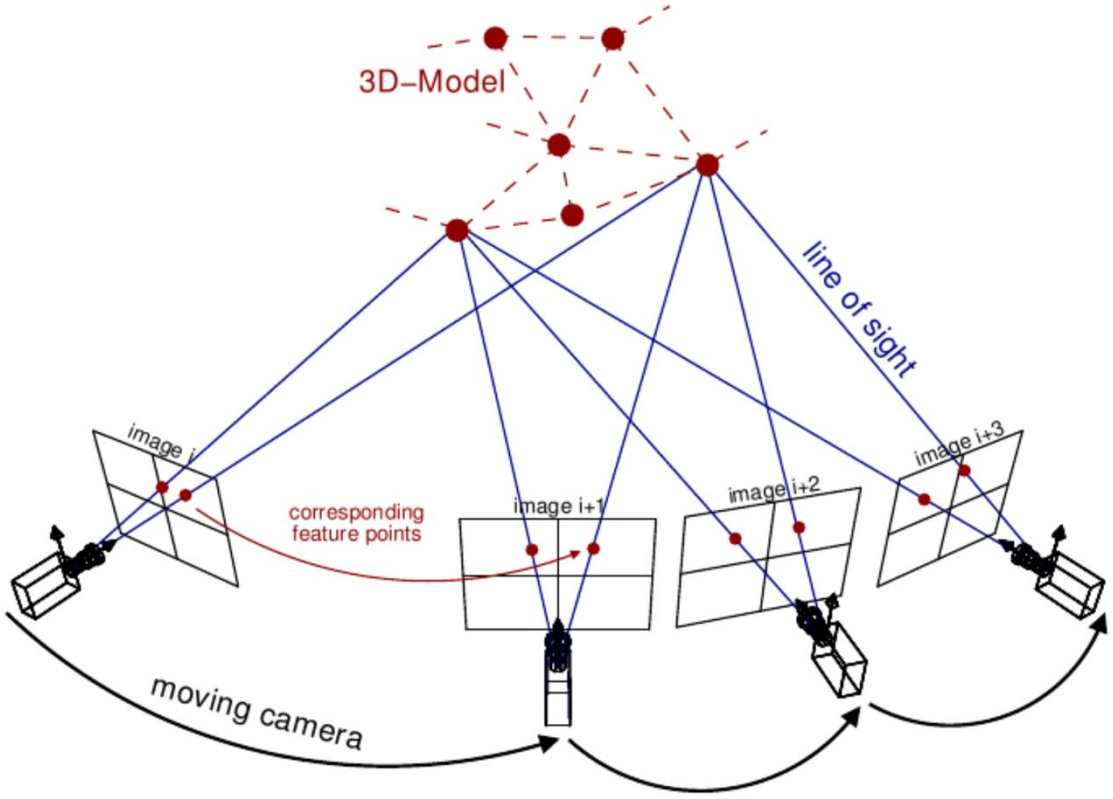
I
Problem Definition

II
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III
Solution

Approach

When reconstructing 3D, we can always merge corresponding points or regions across multi-views.



Schonberger, Johannes L., and Jan-Michael Frahm. "Structure-from-motion revisited." Proceedings of the IEEE conference on computer vision and pattern recognition. 2016.

Approach

When reconstructing 3D, we can always merge corresponding points or regions across multi-views.

How can we merge points or Gaussians modeling similar regions?

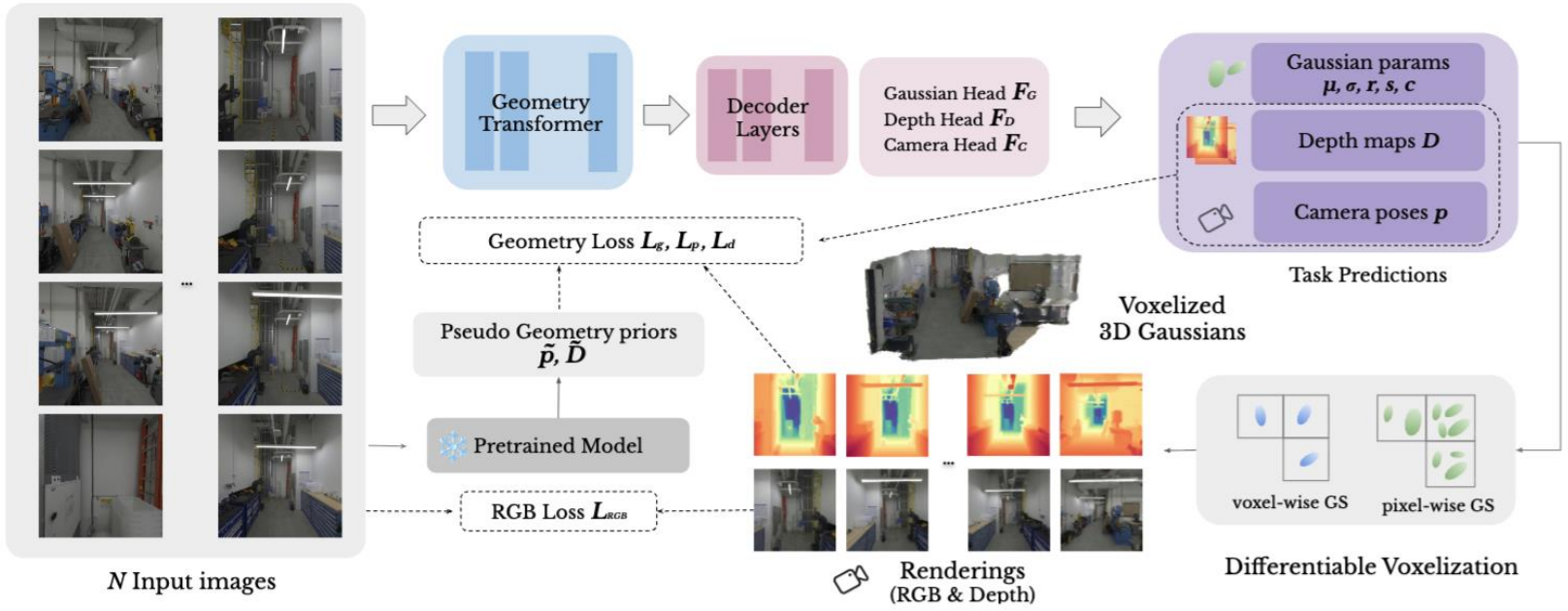
1. *Top-down:* First estimate per-pixel Gaussians -> Merge Gaussians post-hoc
2. *Bottom-up:* First aggregate similar regions -> estimate essential Gaussians only

Approach

When reconstructing 3D, we can always merge corresponding points or regions across multi-views.

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Jiang, Lihan, et al. "AnySplat: Feed-forward 3D Gaussian Splatting from Unconstrained Views." arXiv preprint arXiv:2505.23716 (2025).

Solution

I
Problem Definition

II
Approach

III
Solution

Solution

When reconstructing 3D, we can always merge corresponding points or regions across multi-views.

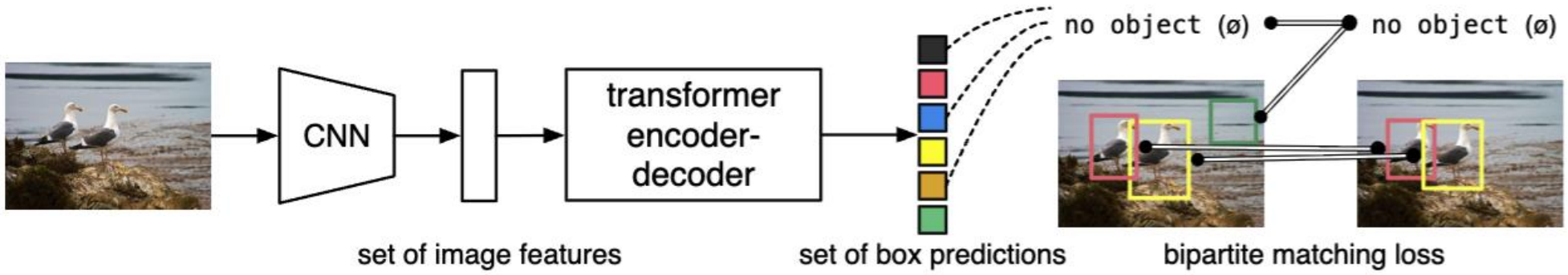
How can we merge points or Gaussians modeling similar regions?

1. *Top-down:* First estimate per-pixel Gaussians -> Merge Gaussians post-hoc
2. *Bottom-up:* First aggregate similar regions -> estimate essential Gaussians only

Solution

How can we estimate only essential Gaussians to reconstruct the scene?

-> We look into transformer decoder structures, which also decode outputs from learna ble queries.

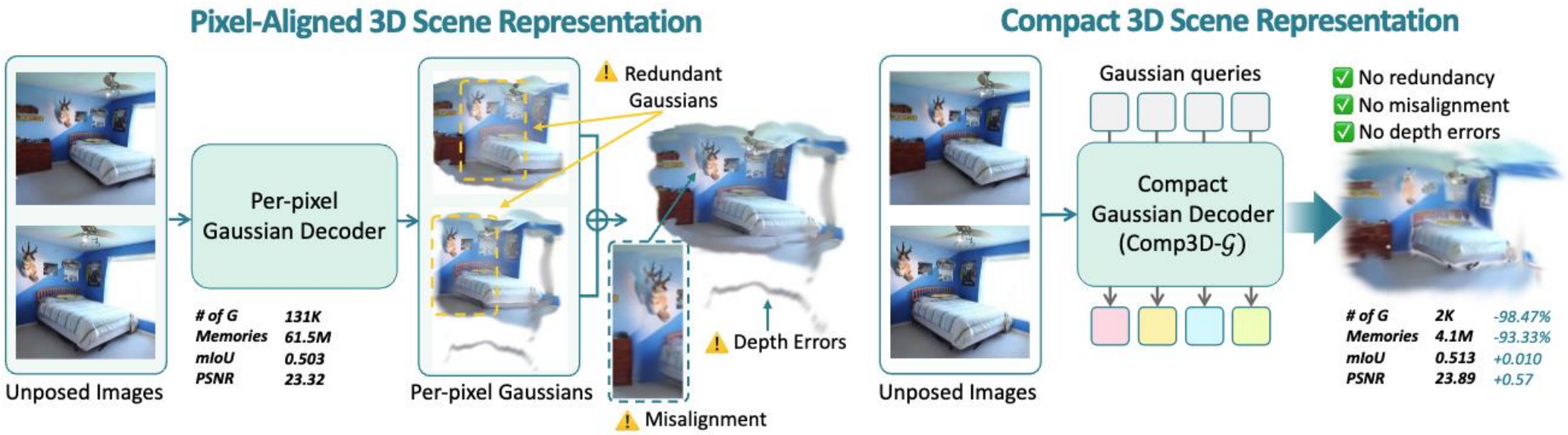


Carion, Nicolas, et al. "End-to-end object detection with transformers." European conference on computer vision. Cham: Springer International Publishing, 2020.

Solution

How can we estimate only essential Gaussians to reconstruct the scene?

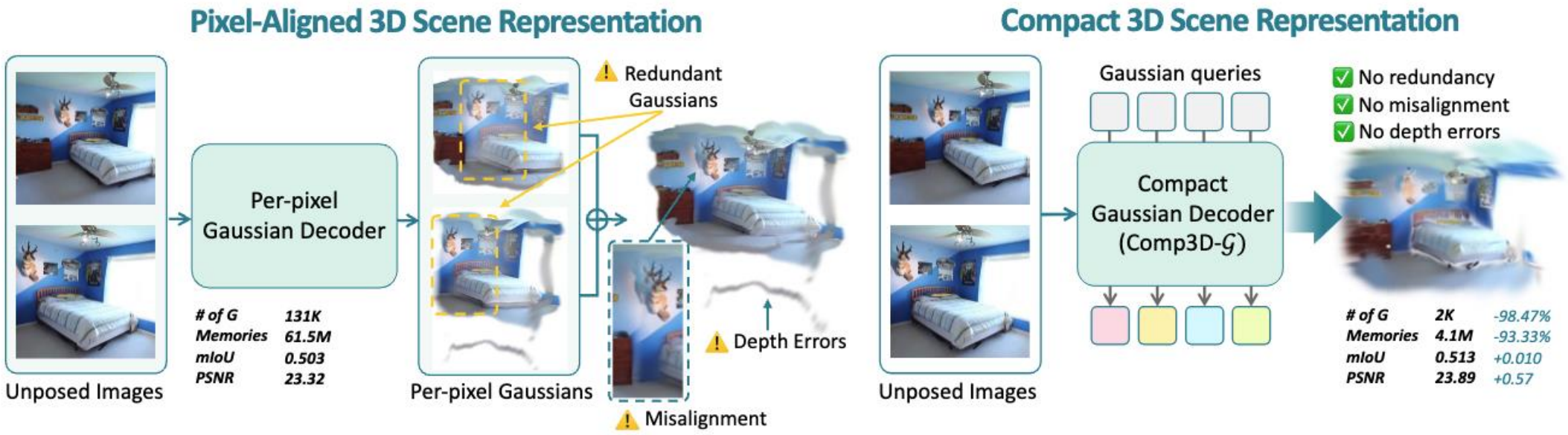
•-> We look into transformer decoder structures, which also decode outputs from learn able queries.



Solution

Compact Gaussian Decoding

1. First extract features from unposed-images (We extract with VGGT, to leverage the l earned priors as in per-pixel approaches)
2. Initialize learnable Gaussian query tokens



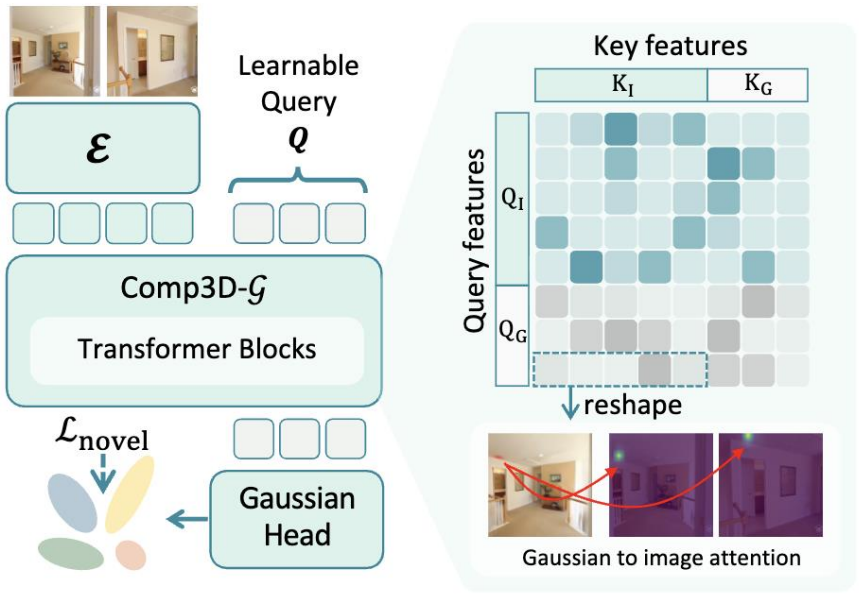
Analysis

Compact Gaussian Decoding

•-> A significant advantage of this approach is that, we can learn the model to estimate essential Gaussian s with only the ***novel view synthesis objective***!

•-> How are these Gaussians learned?

•-> Through ***bi-directional attention*** of the encoder features and learnable queries!



(a) Overall framework



(b) Attention visualization

Analysis

Compact Gaussian Decoding

•-> When handling two 224*224 images as input, per-pixel Gaussian estimators result with 100K Gaussian s.

•-> In contrast, we observe that using only 2K learnable queries, where each query is decoded with a singl e Gaussian is enough

•-> By largely reducing the number of Gaussians, we can now directly lift features to the Gaussians withou t compression.

Analysis

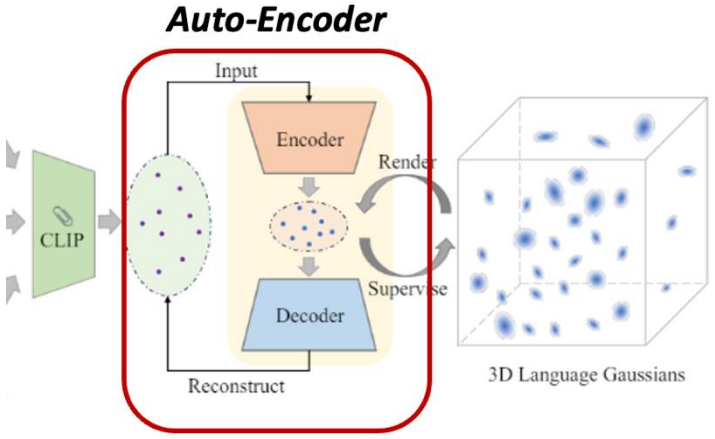
Any-Feature Lifting

What are the existing challenges of feature lifting?

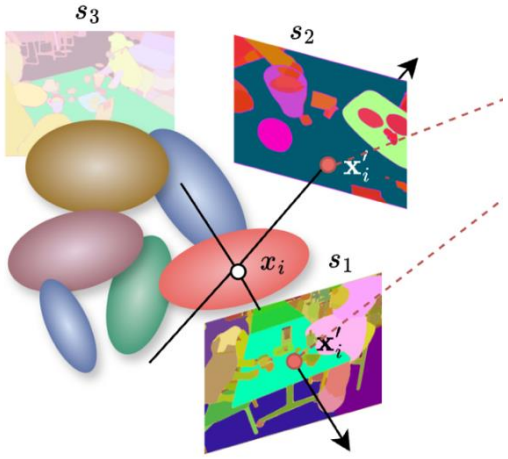
-> Computation: Feature dimensions are too high for rich features (DINOv2 : 768)

•-> 2D-3D Correspondence: We need to identify which Gaussian is projected to which patch

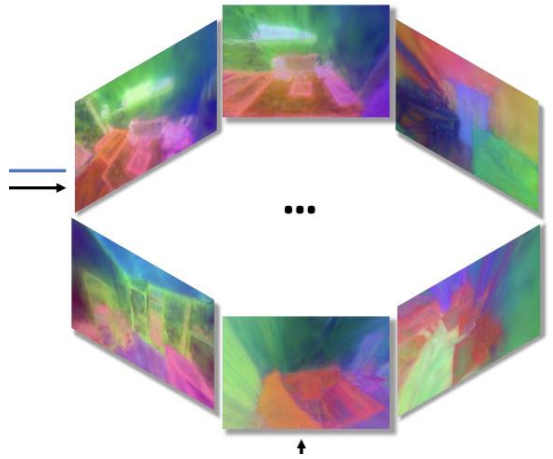
•-> View-inconsistent Features: Feature might be inconsistent across viewpoints



Computation



2D-3D Correspondence



View-Inconsistent Feature

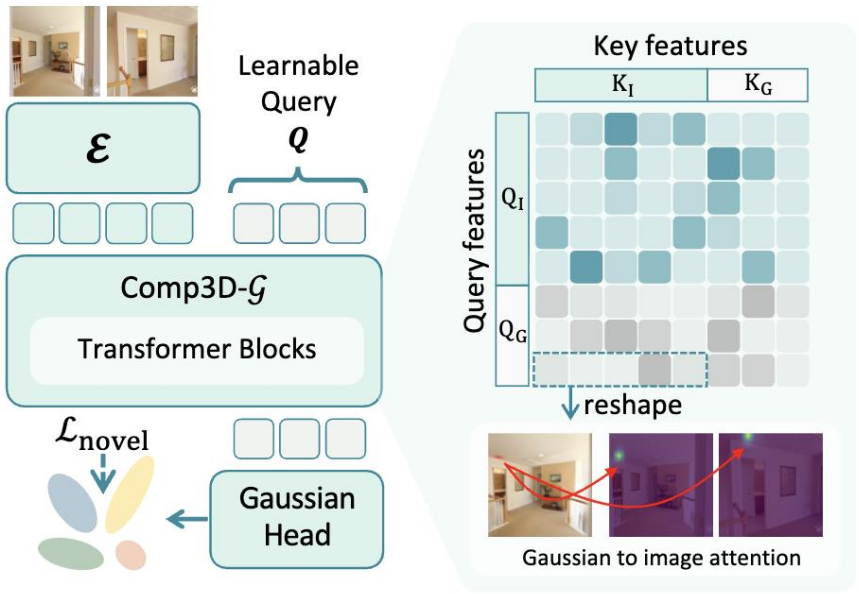
Analysis

Any-Feature Lifting

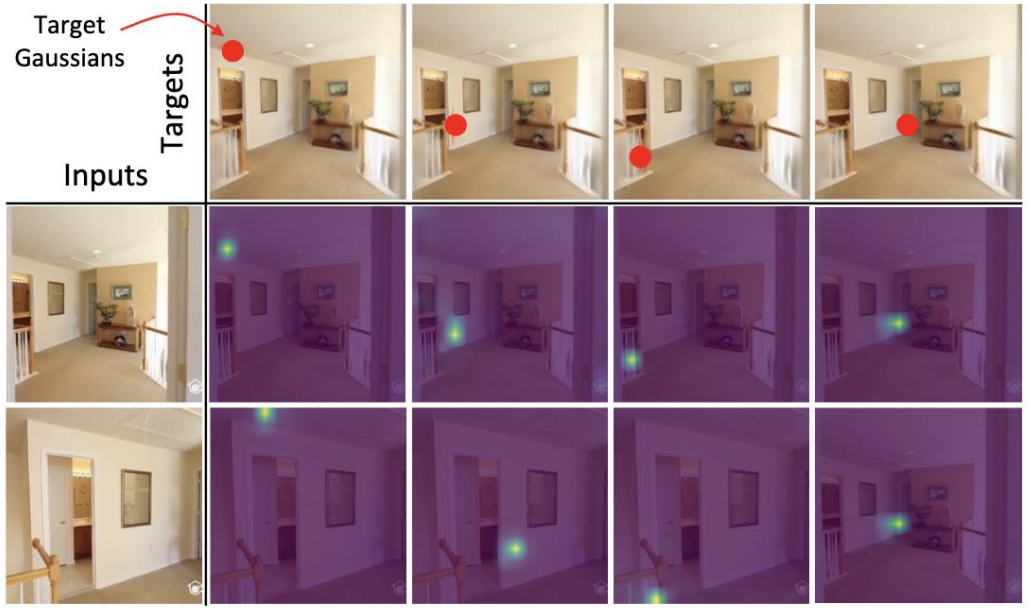
We observe that our trained model can also be used to resolve both problems!

•-> 2D-3D Correspondence: Attention indicates where each Gaussian is looking at!

•-> View-inconsistent Features: Attention weights indicate the aggregation weights!



(a) Overall framework



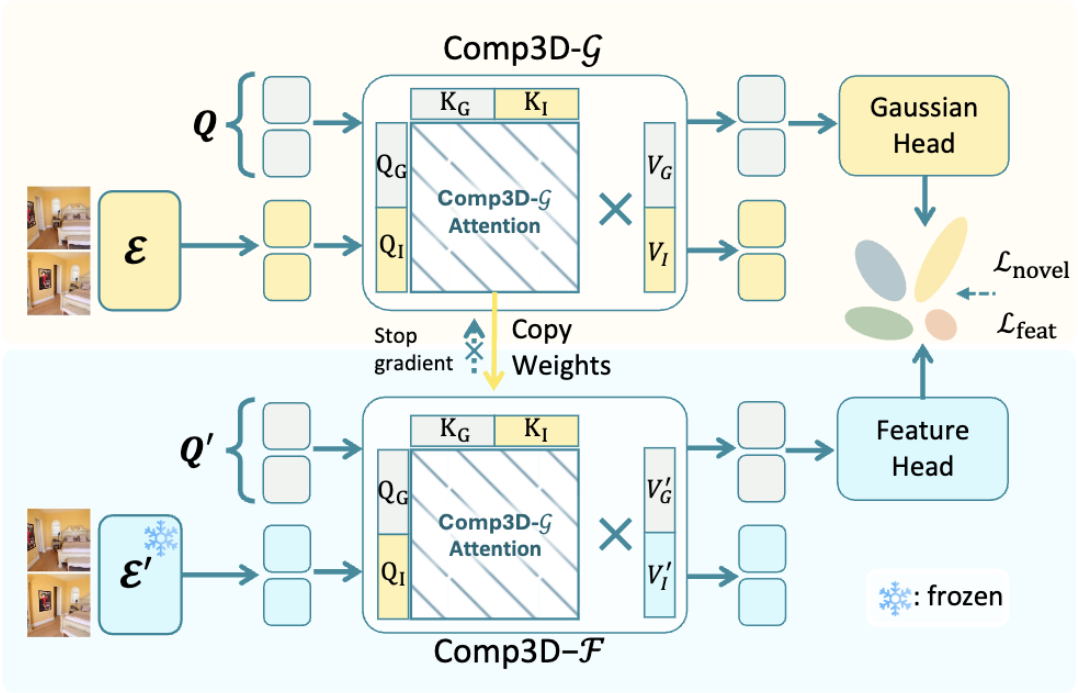
(b) Attention visualization

Analysis

Any-Feature Lifting

We introduce another feature aggregator that decodes the ***view-invariant feature*** to attach to each Gaus sian.

We freeze the attention weights, and introduce new query tokens to handle different dimensions.



Results

Joint Reconstruction and Understanding

(1) Reconstruction: We evaluate the reconstruction quality by the novel view rendering results

(2) 3D Scene Understanding : We evaluate the open-vocabulary segmentation

Table 3. **Comparison of 3D scene understanding on Scannet.** We lift LSeg [23] and MaskCLIP[6] features from two input views and evaluate open-vocabulary segmentation on target views. Our method generates fewer Gaussians while outperforming feed-forward and per-scene optimization methods trained with substantially more posed inputs. ∗: Features directly extracted from target view images.

Target View

Methods	Feature	Feed Forward	Input Pose	LSeg [23] mIoU↑	Acc↑	PSNR↑	SSIM↑	LPIPS↓	MaskCLIP [35] mIoU↑	Acc↑	PSNR↑	SSIM↑	LPIPS↓	#G↓	Memories↓	FPS↑
LSeg / MaskCLIP* [23, 35]	✓	✗	-	0.506	0.797	-	-	-	0.341	0.667	-	-	-	-	-	-
Feature-3DGS [59]	✓	✗	✓	0.379	0.644	19.83	0.684	0.357	0.353	0.663	17.47	0.612	0.420	1,185K	845.2MB	19.2
CF^3 [21]	✓	✗	✓	0.376	0.657	20.04	0.691	0.359	0.336	0.634	20.14	0.695	0.354	53K	38.4MB	252.4
NoPoSplat [53]	✗	✓	✗	-	-	24.59	0.792	0.228	-	-	24.59	0.792	0.228	131K	33.6MB	369.8
LSM [9]	✓	✓	✗	0.503	0.793	23.32	0.767	0.250	0.286	0.505	22.87	0.737	0.286	131K	61.5MB	254.5
Ours	✓	✓	✗	0.513	0.783	23.89	0.770	0.285	0.369	0.675	23.75	0.763	0.290	2K	4.1MB	243.4

Source View

Methods	Feature	Feed Forward	Input Pose	LSeg [23] mIoU↑	Acc↑	PSNR↑	SSIM↑	LPIPS↓	MaskCLIP [35] mIoU↑	Acc↑	PSNR↑	SSIM↑	LPIPS↓	#G↓	Memories↓	FPS↑
LSeg / MaskCLIP* [23, 35]	✓	✗	-	0.521	0.820	-	-	-	0.344	0.665	-	-	-	-	-	-
Feature-3DGS [59]	✓	✗	✓	0.392	0.655	21.73	0.757	0.314	0.353	0.674	22.25	0.777	0.308	1,185K	845.2MB	19.2
CF^3 [21]	✓	✗	✓	0.390	0.668	22.99	0.804	0.272	0.342	0.642	23.16	0.812	0.265	53K	38.4MB	252.4
NoPoSplat [53]	✗	✓	✗	-	-	25.20	0.812	0.217	-	-	25.20	0.812	0.217	131K	33.6MB	369.8
LSM [9]	✓	✓	✗	0.511	0.798	25.44	0.811	0.214	0.251	0.516	25.01	0.824	0.230	131K	61.5MB	254.5
Ours	✓	✓	✗	0.542	0.803	23.92	0.766	0.278	0.361	0.668	23.39	0.759	0.284	2K	4.1MB	243.4

Outline

- About me
- Introduction
- From Spatial AI to Physical AI: Learning to Represent, Reason, and Act in 3D
  - C3G: Learning Compact 3D Representations with 2K Gaussians
    - CVPR 2026 # Compact 3D Gaussian Splatting # 3D Reconstruction # 3D Scene Understanding
  - Imagine3D-LLM: Teaching MLLMs to Imagine 3D Scenes Before Answering
    - NeurIPS 2026 (under review) # Compact 3D tokens # Mental Reconstruction # Spatial Reasoning
  - Geometric Action Model for Visuomotor Control
    - Arxiv 2026 (under review) # WAM # VLA # Manipulation # Geometric Foudnation Model # World Model
- Conclusion
- Future Works

Imagine3D-LLM: Teaching MLLMs to Imagine 3D Scenes Before Answering

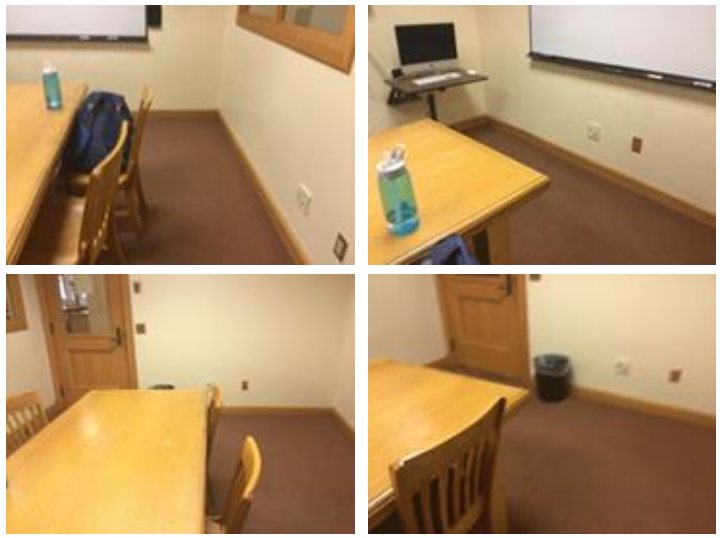
Jaewoo Jung, Hyeonseo Yu, Honggyu An, Jisang Han, Mungyeom Kim, Minkyeong Jeon, Heeseong Shin, WonJun Moon, Federico Tombari, Daniel Barath, Marc Pollefeys, Seungryong Kim, Sunghwan Hong

- NeurIPS - 2026

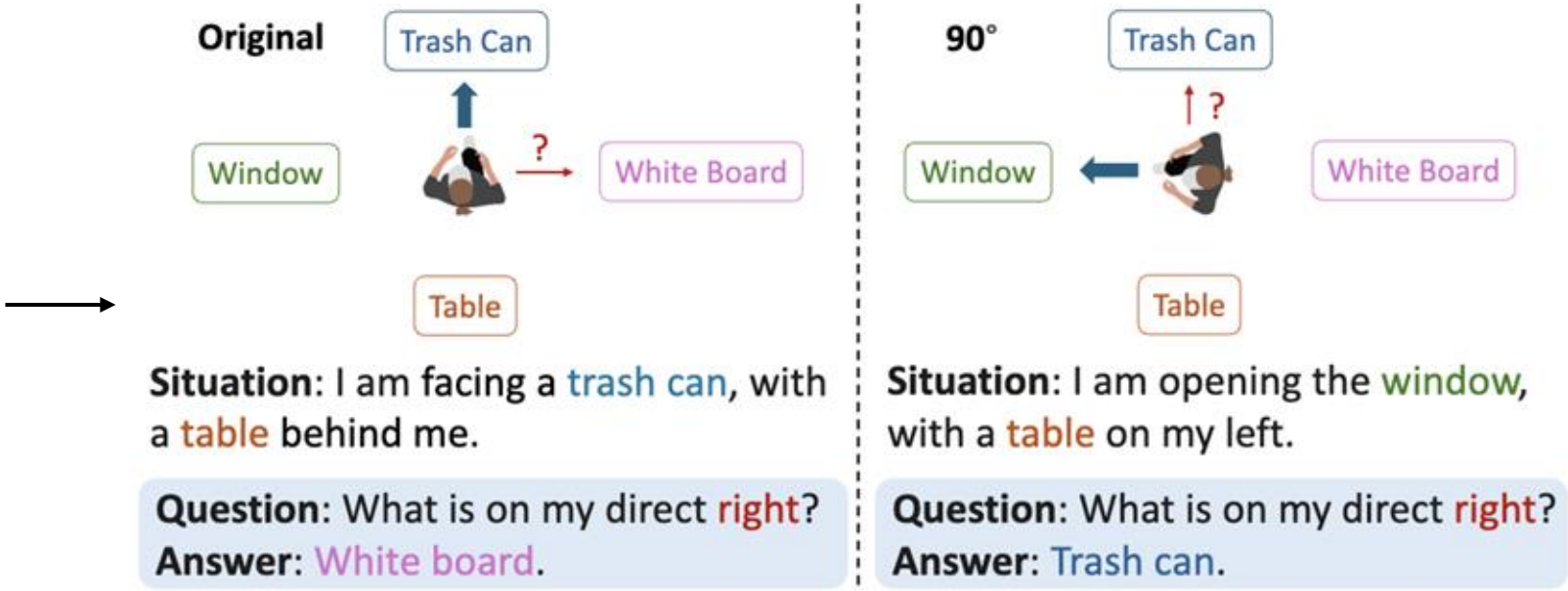
Problem Formulation

Improving the 3D reasoning capabilities of Multimodal Large Language Models

Given multi-view images captured from a scene, we want to leverage the reasoning power of MLLMs to und erstand the 3D scene structure from multi-view images, leading to better QA performance in 3D.



Set of multi-view images



Figures from Ma, Xianzheng, et al. "Do 3D Large Language Models Really Understand 3D Spatial Relationships?." arXiv preprint arXiv:2603.23523 (2026).

Motivation

How can we make MLLMs better understand 3D?

While MLLMs have made significant progress in reasoning and understanding single images, the progress in 3D understanding with multi-view images are relatively slow.

Model	LLM Params	Frames	Date	Avg▼	Numerical Answer Questions				Multiple-Choice Questions			
					Obj. Count	Abs. Dist.	Obj. Size	Room Size	Rel. Dist.	Rel. Dir.	Route Plan	Appear. Order
Baselines												
Chance-level (Random)	-	-	*2024-11-15*	-	-	-	-	-	*25.0*	*36.1*	*28.3*	*25.0*
Chance-level (Frequency)	-	-	*2024-11-15*	**34.0**	*62.1*	*32.0*	*29.9*	*33.1*	*25.1*	*47.9*	*28.4*	*25.2*
Human-Level	-	-	2024-11-15	**79.2**	**94.3**	**47.0**	**60.4**	**45.9**	**94.7**	**95.8**	**95.8**	**100.0**
MLLMs												
SenseNova-SI-1.2-InternVL3-8B OPEN	8B	32	2026-02-04	**69.6**	72.7	56.0	77.1	75.7	70.4	81.7	42.8	79.9
SpaceMind OPEN	7B	32	2025-11-27	**69.2**	72.7	60.3	78.0	75.5	71.0	87.9	42.8	65.9
Gemini-1.5-Pro PROPRIETARY	-	-	2024-11-15	**48.8**	49.6	28.8	58.6	49.4	46.0	48.1	42.0	68.0
InternVL3-78B OPEN	78B	64	2025-04-19	**48.4**	71.2	53.7	44.4	39.5	55.9	39.5	28.9	54.5
GPT-4o PROPRIETARY	-	64	2025-04-03	**47.8**	43.1	34.1	68.6	64.2	48.3	43.1	29.4	51.3

VSI-Bench Leaderboard

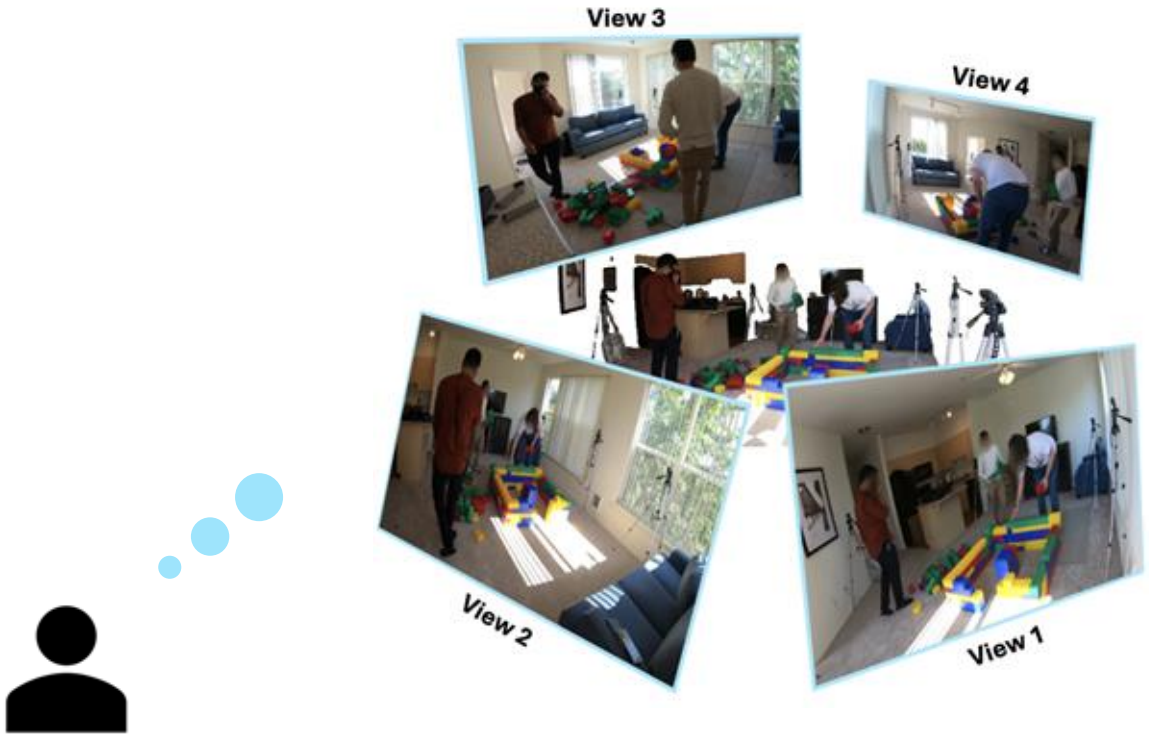
Yang, Jihan, et al. "Thinking in space: How multimodal large language models see, remember, and recall spaces." Proceedings of the Computer Vision and Pattern Recognition Conference. 2025.

Motivation

How can we make MLLMs better understand 3D?

How can we enable machines to better understand the 3D world as we do?

With multi-view images, humans naturally understand how these images shape the scene.

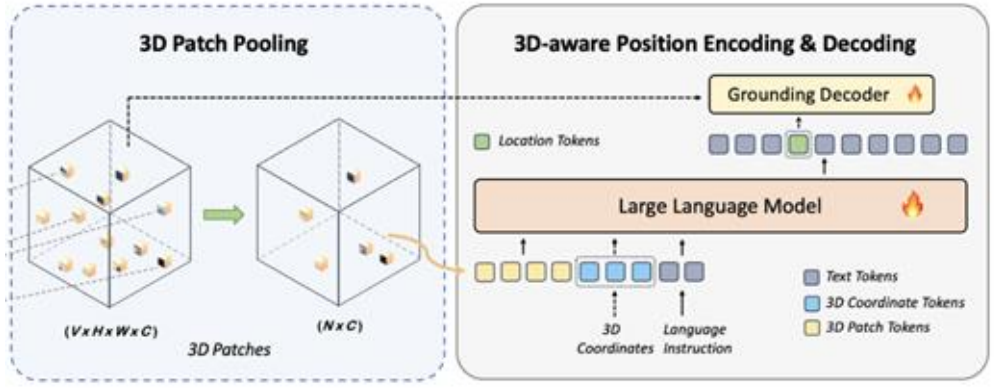


Yeh, Chun-Hsiao, et al. "Seeing from another perspective: Evaluating multi-view understanding in mllms." arXiv preprint arXiv:2504.15280 (2025).

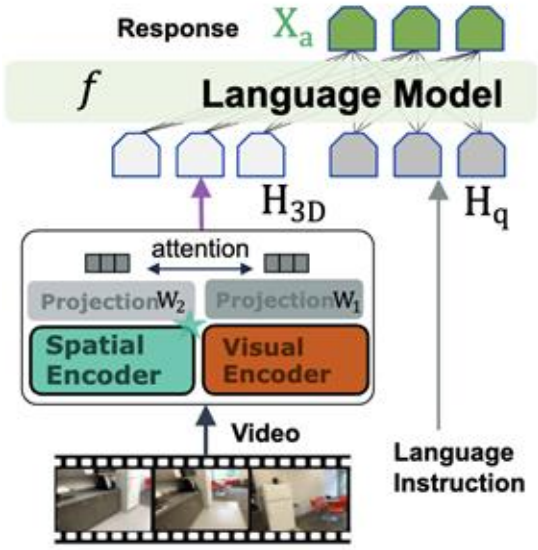
Motivation

How can we make MLLMs better understand 3D?

Recent approaches either: (1) improve the image features to capture better correspondence, (2) fuse image features with recent geometric foundation models (e.g., Cut3r, VGGT).



Correspondence boosting



Fusion with geometric foundation models

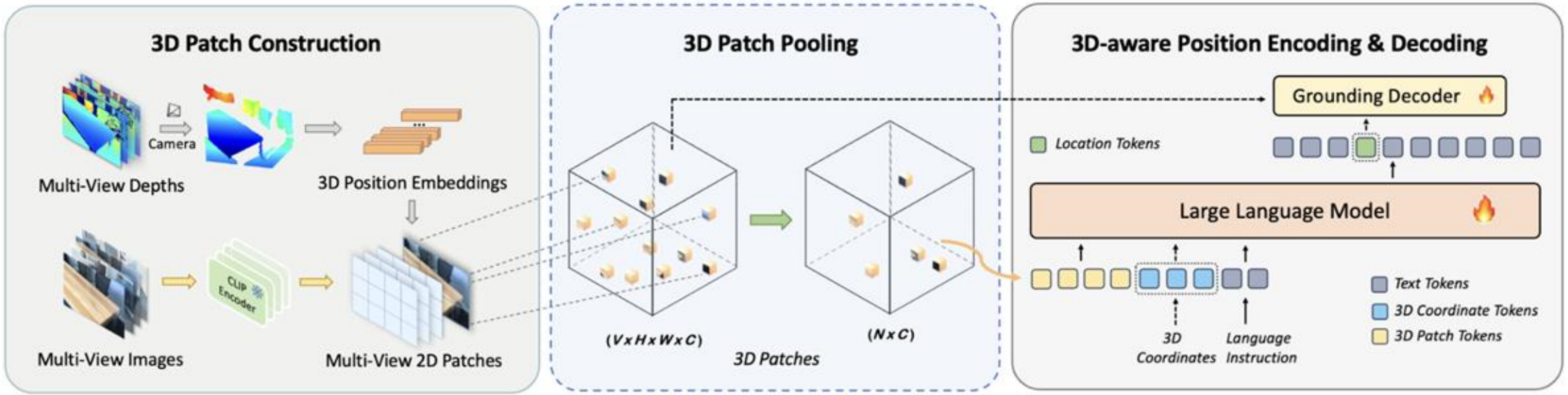
(Left): Zhu, Chenming, et al. "Llava-3d: A simple yet effective pathway to empowering lmms with 3d-awareness." arXiv preprint arXiv:2409.18125 (2024).
(Right): Fan, Zhiwen, et al. "Vlm-3r: Vision-language models augmented with instruction-aligned 3d reconstruction." arXiv preprint arXiv:2505.20279 (2025).

Motivation

How can we make MLLMs better understand 3D?

Recent approaches enable multi-view understanding by adding the ***3D-aware Position Encoding***, which em beds the visual tokens with the corresponding 3D coordinates.

-> These works underscore the importance of teaching MLLMs to understand correspondence!

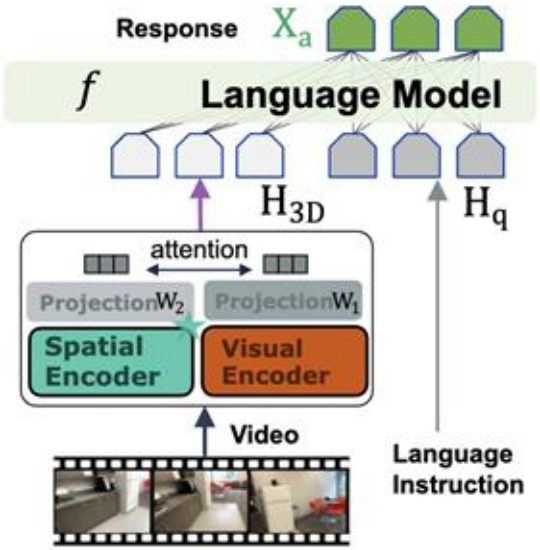


Zhu, Chenming, et al. "Llava-3d: A simple yet effective pathway to empowering lmms with 3d-awareness." arXiv preprint arXiv:2409.18125 (2024).

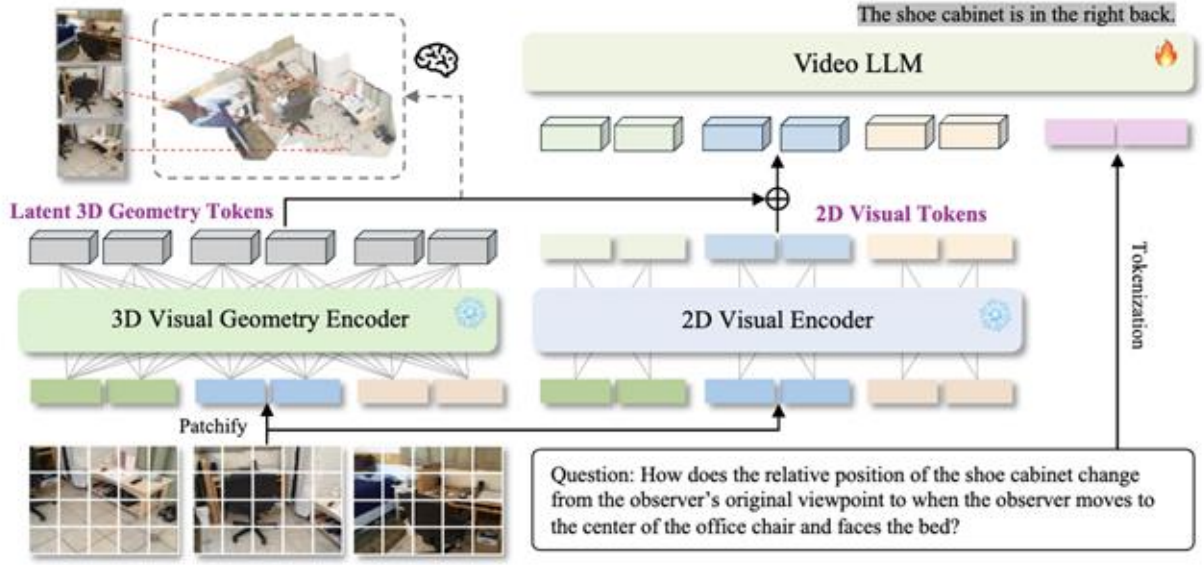
Motivation

How can we make MLLMs better understand 3D?

Recent approaches also introduce a fusion layer with geometric foundation features either in the input laye r or middle layers of the LLM.



VLM-3R



VG-LLM

(Left): Fan, Zhiwen, et al. "Vlm-3r: Vision-language models augmented with instruction-aligned 3d reconstruction." arXiv preprint arXiv:2505.20279 (2025).
(Right): Zheng, Duo, et al. "Learning from videos for 3d world: Enhancing mllms with 3d vision geometry priors." arXiv preprint arXiv:2505.24625 (2025).

Motivation

How can we make MLLMs better understand 3D?

However, merely boosting pixel-level correspondence or fusing with foundation features capable of estimati ng pixel-level geometry still shows large gaps compared to human reasoning.

-> Is this the right direction?

As humans, rather than modeling pixel-level correspondences, we easily identify common objects appearin g in the scene, where we can then imagine how the scene is structured through a process we call "***Mental R econstruction***".



Qi, Zhangyang, et al. "Gpt4scene: Understand 3d scenes from videos with vision-language models." arXiv preprint arXiv:2501.01428 (2025).

Motivation

How can we make MLLMs better understand 3D?

In this work, we make MLLMs mimic the behavior of humans by enabling the MLLMs to identify common o bjects across views and understanding coarse 3D structure before answering 3D questions.



Qi, Zhangyang, et al. "Gpt4scene: Understand 3d scenes from videos with vision-language models." arXiv preprint arXiv:2501.01428 (2025).

Architecture

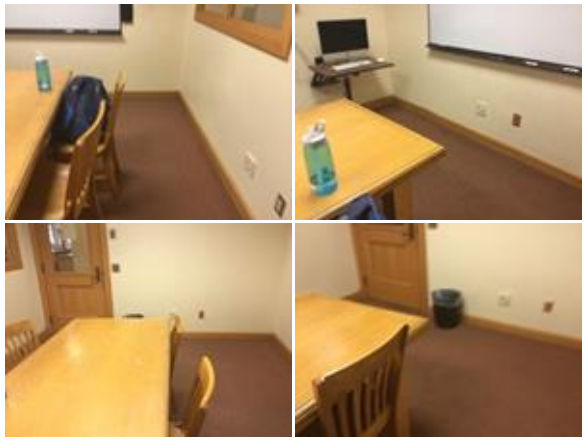
How can we incorporate such mental reconstruction capability to MLLMs?

To achieve this we want the MLLMs to learn the following behavior:

(i) Learn to identify common objects in multi-view images

(ii) Based on these objects, learn how the scene is coarsely structured.

-> We do not need perfect reconstruction performance, but we just want the model to learn how to map im ages to 3D.



Coarse-level
Understanding is enough!

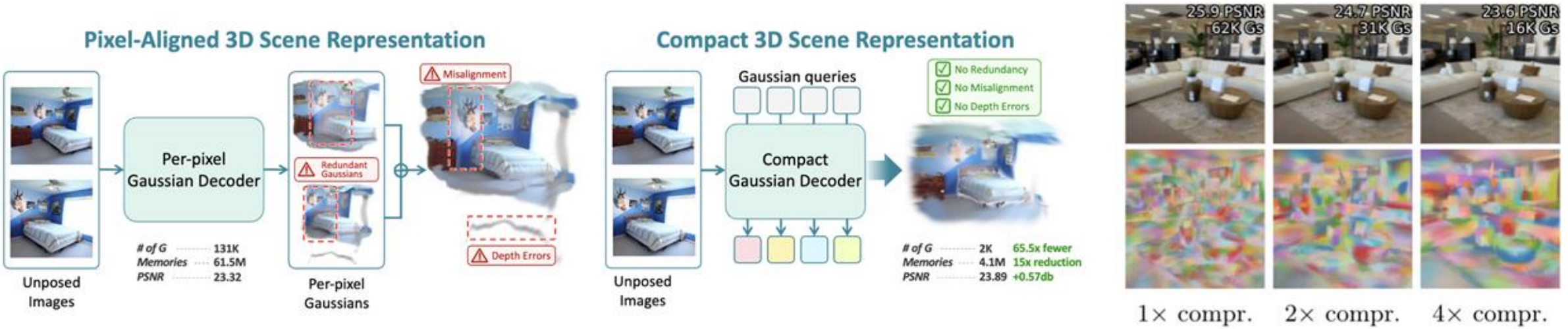


Architecture

How can we incorporate such mental reconstruction capability to MLLMs?

How can we make a model learn such abstract reconstructions?

Extend ideas from recent approaches in feed-forward 3DGS estimation done through initializing queries to estimate Gaussians instead of doing pixel-level 3DGS estimation.



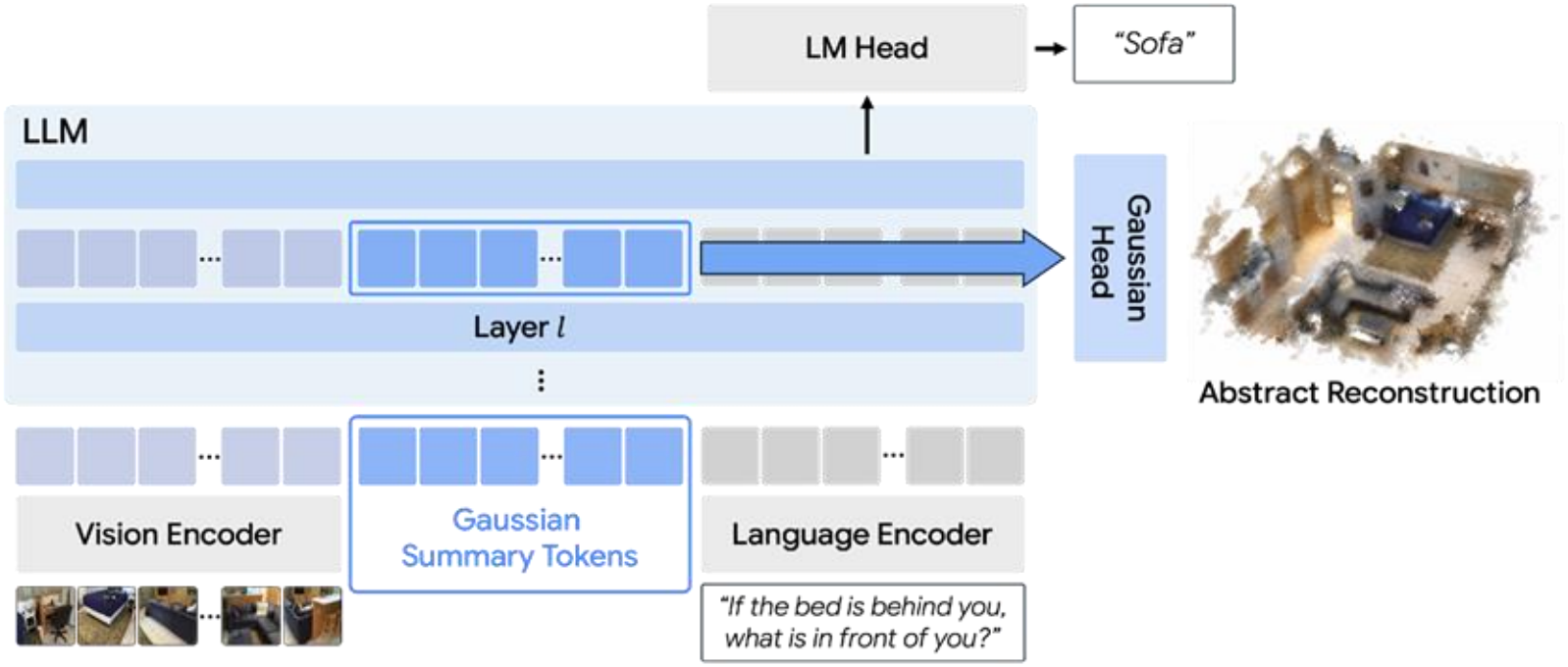
An, Honggyu, Jung*, Jaewoo et al. "C3G: Learning Compact 3D Representations with 2K Gaussians." CVPR 2026.*

Architecture

How can we incorporate such mental reconstruction capability to MLLMs?

We introduce a set of learnable Gaussian summary tokens that are appended after the image tokens, wher e these tokens are then further decoded to 3DGS parameters with a shallow Gaussian head.

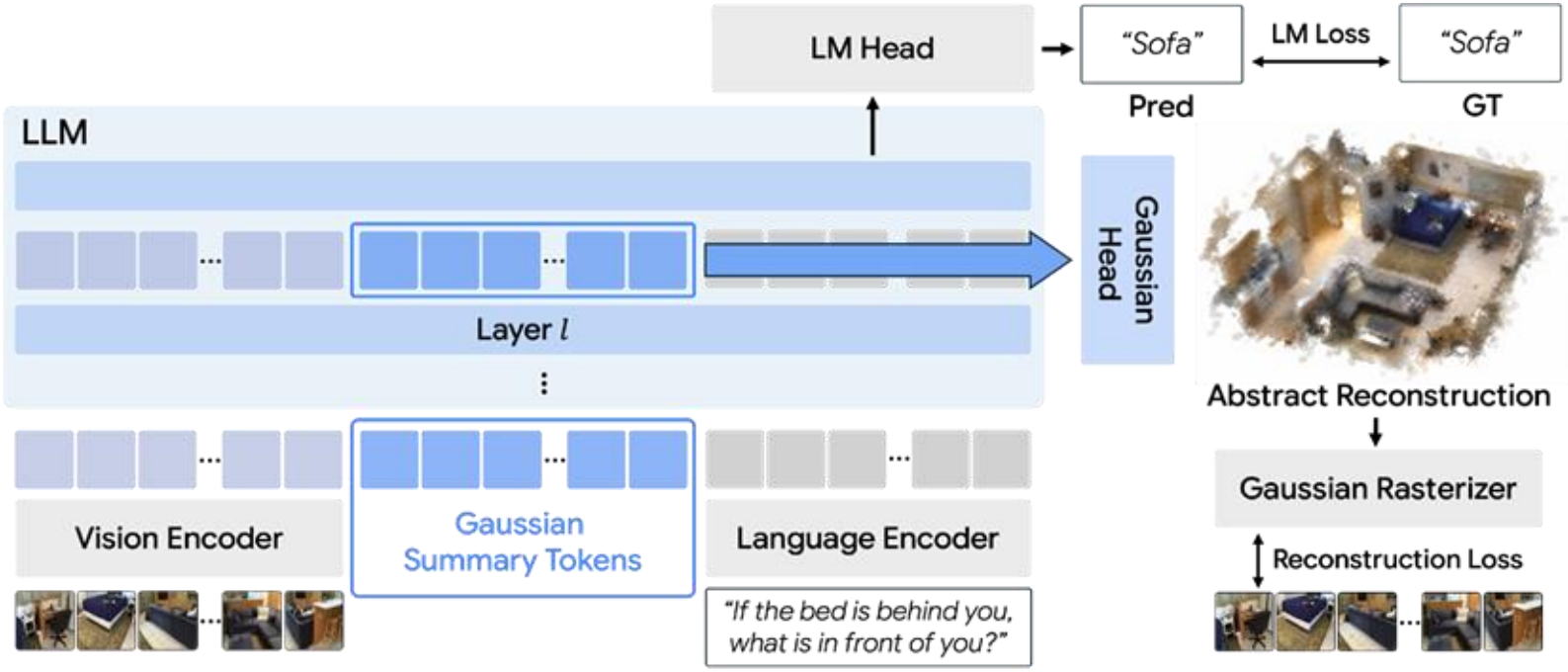
-> By defining **fewer** number of summary tokens compared to image tokens, the model must learn to map c ommon objects to the same token to successfully reconstruct the scene.



Architecture

How can we incorporate such mental reconstruction capability to MLLMs?

The model is trained jointly with the novel view synthesis loss L_{NVS}(L1, Lpips, SSIM) and the traditional next token prediction loss L_{NTP}.

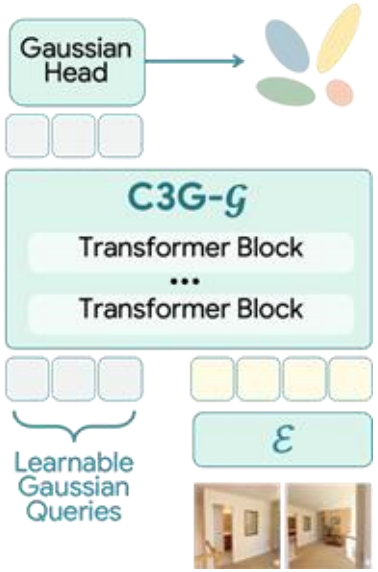


Experiments

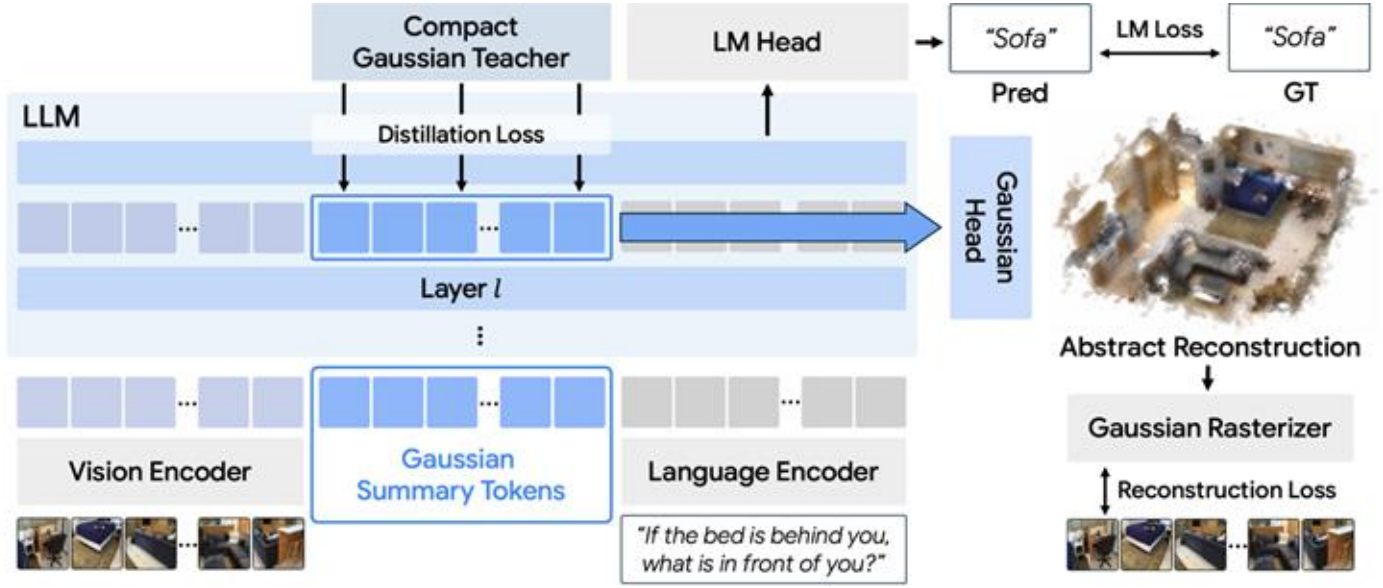
How can we easily train such framework?

The training of feed-forward 3DGS is known to require (1) long training iterations with (2) large batch sizes, requiring substantially more iterations than is practical given the cost of MLLM finetuning.

-> To enable training with fewer iterations, we adopt additional supervision of (1) token feature distillation a nd (2) Gaussian parameter supervision obtained from a Compact Gaussian Teacher.



Compact Gaussian Teacher



Analysis and Results

Does learning abstract reconstruction help?

After training on the combination of 3DQA datasets, our method achieves state-of-the-art performance in multiple benchmarks.

Table 1: **Quantitative results on 3D question-answering** over SQA3D [51], Real-3DQA [50], and ScanQA [3]. "Specialists" refer to models tailored to individual tasks via task-specific decoders. General-purpose 2D LMMs [65, 72, 86] are reported under a zero-shot protocol. We report values available from prior works; "–" marks entries unavailable to us.

Method	$SQA3D_{test}$		$Real\text{-}3DQA_{test}$		$ScanQA_{val}$				
	EM	EM-R	EM	EM-R	CIDEr	BLEU-4	METEOR	ROUGE	EM
Specialists									
SQA3D [51]	46.6	–	–	–	–	–	–	–	–
ScanQA [3]	–	–	–	–	64.9	10.1	13.1	33.3	21.1
2D LMMs									
InternVL2-8B [65]	33.0	45.3	–	–	62.5	3.3	14.5	34.3	–
Qwen2-VL-7B [72]	40.7	46.7	–	–	53.9	3.0	11.4	29.3	–
LLaVA-Video-7B [86]	48.5	–	–	–	88.7	3.1	17.7	44.6	–
3D LMMs									
Scene-LLM [24]	53.6	–	–	–	80.0	11.7	15.8	35.9	27.2
LL3DA [13]	–	–	–	–	76.8	–	15.9	37.3	–
LEO [30]	50.0	52.4	–	–	80.0	11.5	16.2	39.3	21.5
ChatScene [29]	54.6	57.5	–	–	87.7	14.3	18.0	41.6	21.6
Grounded 3D-LLM [15]	–	–	–	–	72.7	13.4	–	–	–
LLaVA-3D [93]	55.6	57.6	–	–	91.7	14.5	20.7	50.1	27.0
Video-3D-LLM [90]	58.6	–	31.2	36.4	102.1	16.4	20.0	49.3	30.1
3DRS [31]	60.6	–	-	-	104.8	-	-	-	30.3
Ross3D [70]	63.0	65.7	36.6	41.5	107.0	17.9	20.9	**50.7**	**30.8**
Imagine3D-LLM (Ours)	**63.8**	**66.4**	**39.2**	**44.3**	**109.3**	**20.5**	**21.3**	49.7	29.9

Analysis and Results

Does learning abstract reconstruction help?

After training on the combination of 3DQA datasets, our method achieves state-of-the-art performance in multiple benchmarks.

Table 2: **Evaluation of 3D dense captioning and visual grounding** on Scan2Cap [11], ScanRefer [10], and Multi3DRefer [85].

Method	$\text{Scan2Cap}_{\text{val}}$ (IoU@0.5)				$\text{ScanRefer}_{\text{val}}$		$\text{Multi3DRefer}_{\text{val}}$	
	ROUGE	BLEU-4	METEOR	CIDEr	Acc@0.25	Acc@0.5	F1@0.25	F1@0.5
Specialists								
Scan2Cap [11]	44.5	23.3	22.0	35.2	–	–	–	–
3DJCG [8]	50.8	31.0	24.2	49.5	49.6	37.3	–	26.6
ScanRefer [10]	–	–	–	–	37.3	24.3	–	–
M3DRef-CLIP [85]	–	–	–	–	51.9	44.7	42.8	38.4
3D LMMs								
LL3DA [13]	55.1	36.8	26.0	65.2	–	–	–	–
Ground 3D-LLM [15]	–	–	–	–	47.9	44.1	45.2	40.6
LEO [30]	58.1	38.2	27.9	72.4	–	–	–	–
ChatScene [29]	58.1	36.3	–	77.1	55.5	50.2	57.1	52.4
LLaVA-3D [93]	63.4	41.1	30.2	79.2	54.1	42.4	–	–
Video-3D-LLM [90]	62.3	42.4	28.9	83.8	58.1	51.7	58.0	52.7
VG-LLM [89]	62.6	41.5	28.9	80.0	-	-	-	-
3DRS [31]	-	41.6	-	86.1	-	-	-	-
Ross3D [70]	66.9	43.4	30.3	81.3	61.1	54.4	59.6	54.3
Imagine3D-LLM (Ours)	**67.6**	**48.3**	**32.6**	**99.2**	**62.8**	**56.3**	**60.2**	**55.0**

Table 3: **Evaluation on SPAR-Bench** [84].

Model	SPAR-Bench			
	Avg.	Low	Med.	High
Proprietary				
GPT-4o [32]	36.4	29.3	24.9	45.1
Claude-3.7-Sonnet [2]	21.8	25.4	7.3	23.3
General-purpose 2D LMMs				
LLaVA-Video-7B [86]	32.3	23.6	24.8	42.6
Qwen2-VL-7B [72]	30.7	27.5	20.4	37.0
InternVL2.5-8B [18]	36.3	29.5	31.9	43.8
Qwen2.5-VL-72B [4]	39.4	35.4	23.1	48.4
Spatially-aware				
VLM3R-7B [22]	43.2	39.8	28.4	51.2
G^2VLM-SR-2B [28]	54.9	60.0	36.3	56.5
3DThinker-7B [17]	63.3	-	-	-
Imagine3D-LLM-7B (Ours)	**68.5**	**60.5**	**67.0**	**76.0**

Analysis and Results

How does learning abstract reconstruction help?

We find that our method improves the overall feature representations of the image tokens. Compared to th e baseline, the PCA visualizations of the image features show much more cleaner feature maps, with corres ponding objects across viewpoints show more similar features.



Baseline (Video-3D-LLM)

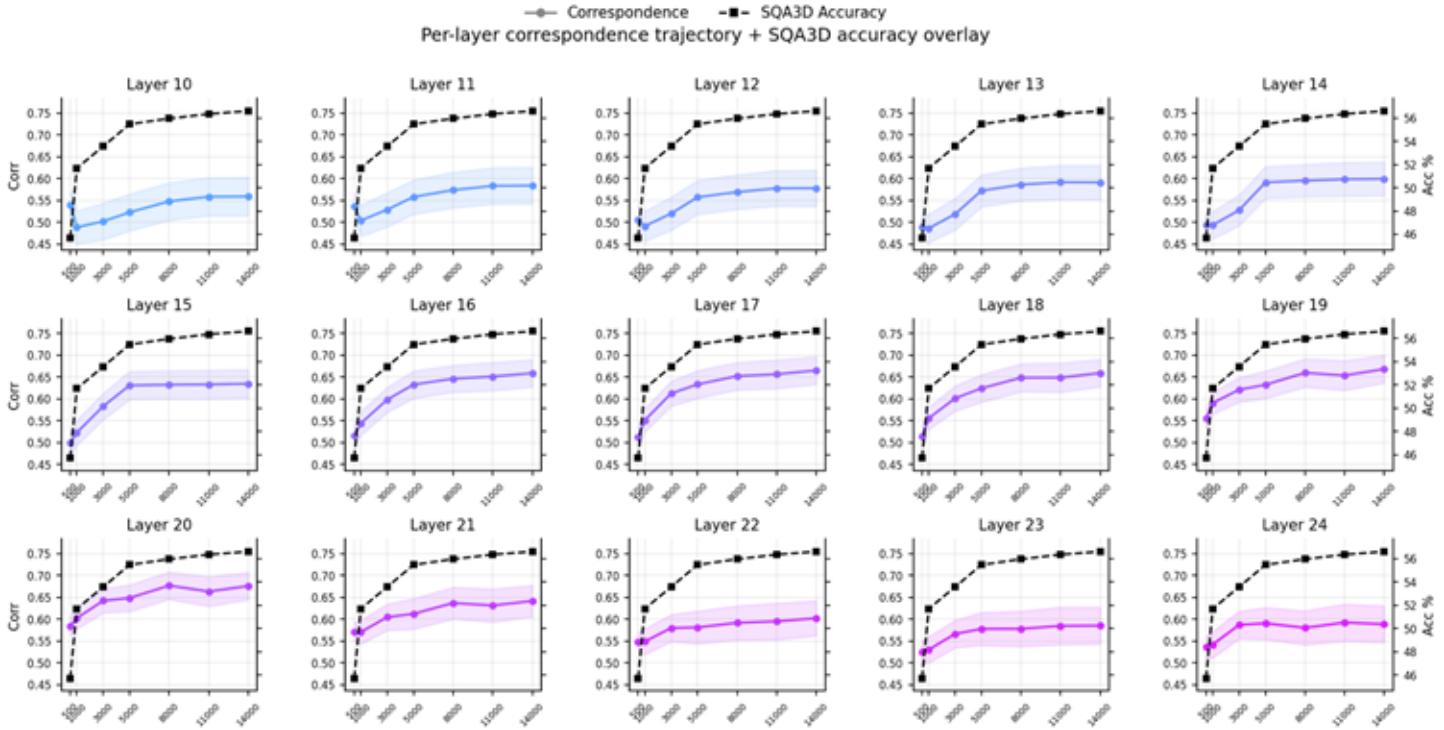


Ours

Analysis and Results

How does learning abstract reconstruction help?

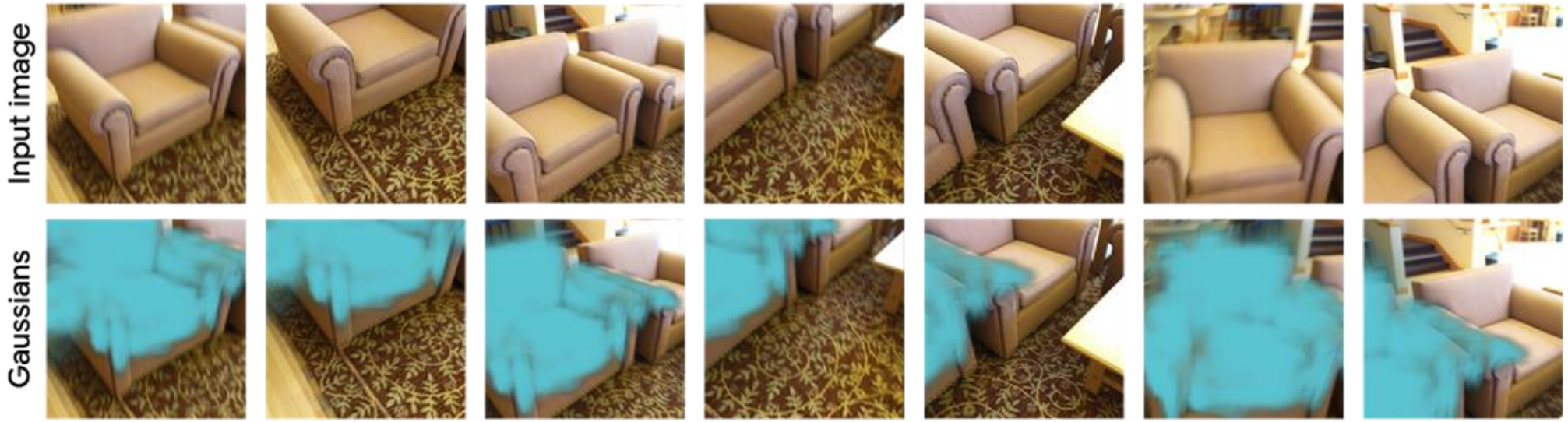
Similar to our intuition, by learning to reconstruct the scene with the Gaussian summary tokens, the model learns to map similar regions to the same token, resulting in boosting the correspondence between the ima ge features as training iterates.



Analysis and Results

What does the Gaussian summary tokens learn to represent?

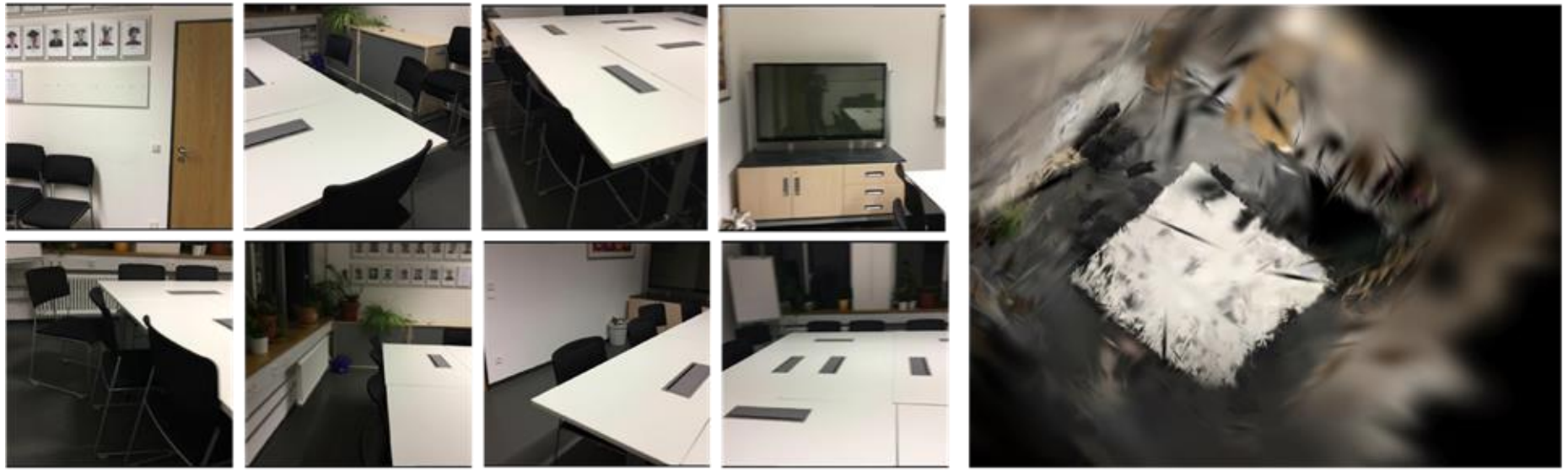
After training, to estimate 3DGS that best represents the scene from limited number of tokens, the model l earns to map similar regions to the same token. To analyze this behavior, we apply K-means to the Gaussian summary tokens and visualize the Gaussian estimated from the same cluster. We observe that similar sum mary tokens are mapped to the same object without any explicit clustering supervision.



Analysis

How well does the model learn to estimate Gaussians?

Also similar to our intuition, the model learns to estimate Gaussians in an coarse and abstract level, with a l ot of parts noisy. We believe that reconstruction quality does not have to pixel-perfect for understanding.



Input Images

Reconstructed Gaussians

Analysis

How well does the model learn to estimate Gaussians?

While increasing the number of Gaussian summary tokens improves reconstruction quality, we find that inc reasing the number of summary tokens weakens the bottleneck behavior, which is more important for the model to understand which objects should be mapped to a single token. Similarly having to few also harms performance where we choose 80 tokens per image as an ideal number.

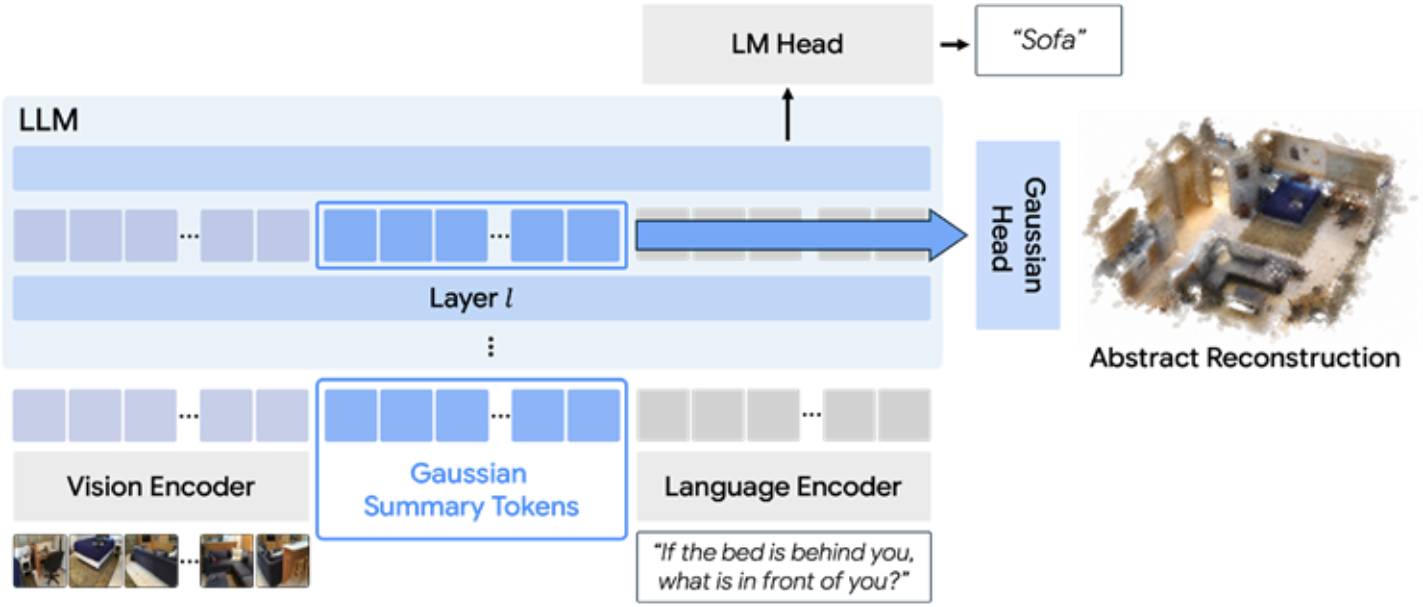


Table 6: **Number of Gaussian summary tokens M per image.**

M	$SQA3D_{test}$	SPAR-Bench
40	61.9	64.6
80 (Ours)	**63.8**	**68.5**
160	60.3	58.4

Outline

- About me
- Introduction
- From Spatial AI to Physical AI: Learning to Represent, Reason, and Act in 3D
  - C3G: Learning Compact 3D Representations with 2K Gaussians
    - CVPR 2026 # Compact 3D Gaussian Splatting # 3D Reconstruction # 3D Scene Understanding
  - Imagine3D-LLM: Teaching MLLMs to Imagine 3D Scenes Before Answering
    - NeurIPS 2026 (under review) # Compact 3D tokens # Mental Reconstruction # Spatial Reasoning
  - Geometric Action Model for Visuomotor Control
    - Arxiv 2026 (under review) # WAM # VLA # Manipulation # Geometric Foudnation Model # World Model
- Conclusion
- Future Works

Geometric Action Model for Visuomotor Control

Jisang Han*, Seonghu Jeon*, Jaewoo Jung, René Zurbrügg, Honggyu An, Tifanny Portela, Marco Hutter, Marc Pollefeys, Seungryong Kim, Sunghwan Hong

- Under Review - 2026

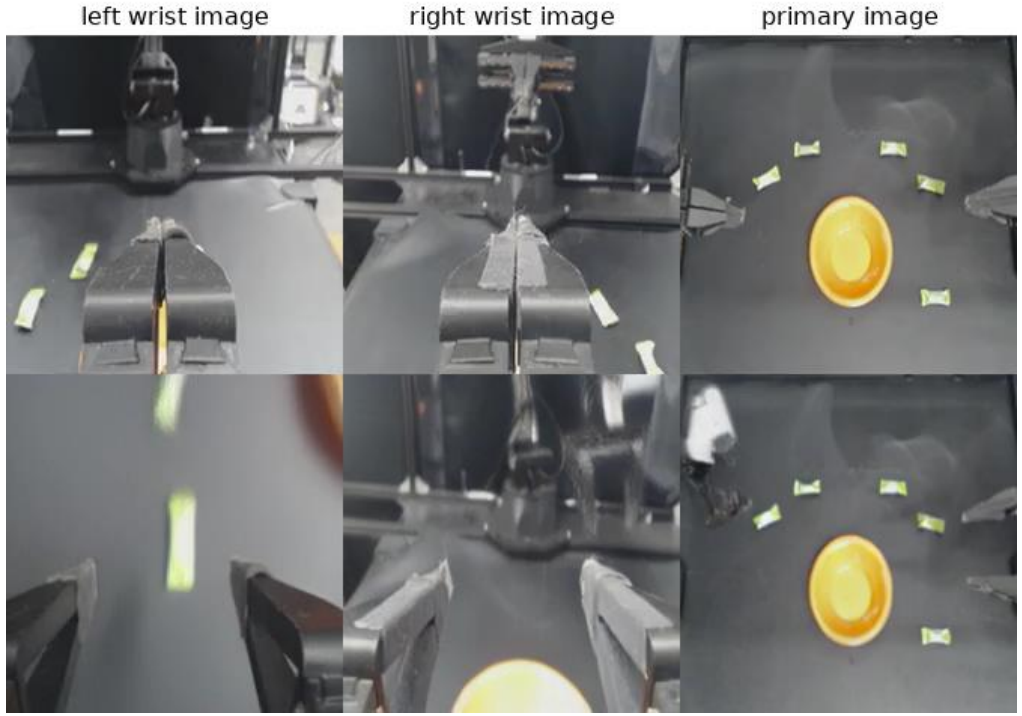
Outline

1. Task Definition
2. Motivation & Related Work
3. Method
4. Experiments
5. Analysis and Discussion

Language Conditioned Robot Manipulation

Langauge conditioned Robot manipulation

- **Objective: execute a natural-language instruction by changing object states through robot actions: pick, place, stack, insert, open, …**
- **Observation: multi-view RGB cameras and proprioception**



"Put candies in bowl"

Cosmos Policy: Fine-Tuning Video Models for Visuomotor Control and Planning — Cosmos Lab

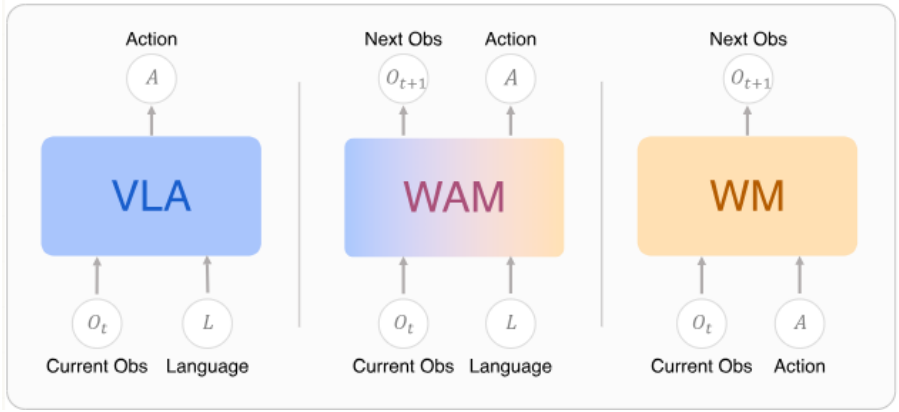
Existing Approaches

Vision-Language-Action Models (VLAs)

- Based on pretrained VLMs, then finetuned for robot action prediction
- Strength: language grounding and open-vocabulary visual recognition
- Problem: backbone itself does not trained for temporal / 3d understanding

World-Action models (WAMs)

- Based on pretrained video or world models, then adapted to predict future states and actions
- Strength: temporal prediction and action-conditioned dynamics
- Problem: too slow inference, still lack of 3d understanding



Limitations of Current Approaches

Strong semantic or temporal priors still leave *geometry* implicit.

- **VLMs and WAMs are trained fundamentally on 2D;**
  action decoder should disentangle 3D cues (depth, scale, occlusion) alone
- **2D VLMs struggles across environment changes, such as camera viewpoint change**

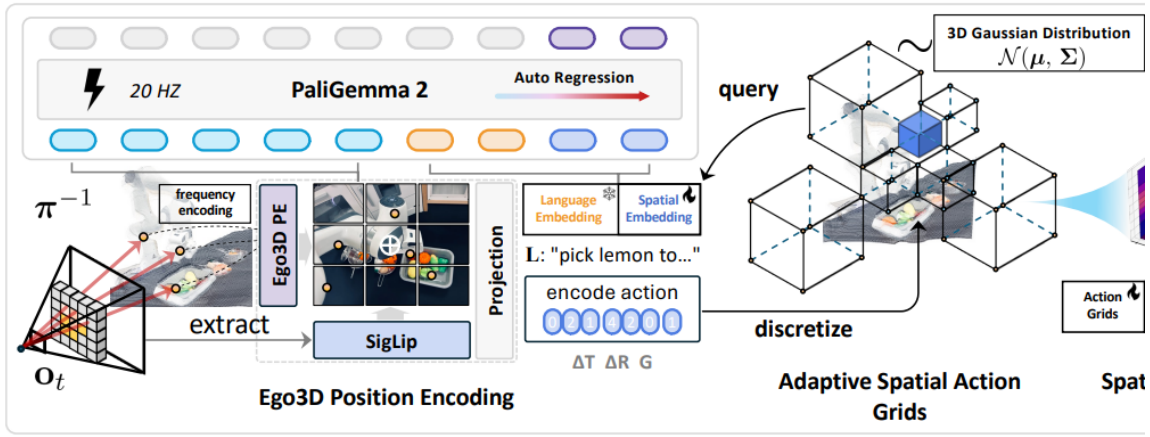
	Original	Camera	Robot	Language	Light	Background	Noise	Layout
OpenVLA	76.5	1.1 ↓75.4	4.1 ↓72.4	26.8 ↓49.7	4.4 ↓72.1	25.3 ↓51.2	19.3 ↓57.2	31.6 ↓44.9
OpenVLA-OFT	97.1	59.7 ↓37.4	37.2 ↓59.9	81.5 ↓15.6	85.8 ↓11.3	92.4 ↓4.7	76.7 ↓20.4	77.1 ↓20.0
OpenVLA-OFT_w	95.3	16.8 ↓78.5	43.7 ↓51.6	73.2 ↓22.1	68.2 ↓27.1	92.5 ↓2.8	51.4 ↓43.9	72.3 ↓23.0
OpenVLA-OFT_m	97.6	57.9 ↓39.7	30.6 ↓67.0	83.6 ↓14.0	91.6 ↓6.0	83.6 ↓14.0	76.3 ↓21.3	73.2 ↓24.4
π_0	94.2	15.8 ↓78.4	6.6 ↓87.6	61.0 ↓33.2	79.6 ↓14.6	78.5 ↓15.7	79.4 ↓14.8	70.4 ↓23.8
π_0-fast	85.5	66.4 ↓19.1	24.8 ↓60.7	63.3 ↓22.2	73.0 ↓12.5	67.7 ↓17.8	75.8 ↓9.7	70.3 ↓15.2
Nora	87.9	4.0 ↓83.9	41.1 ↓46.8	67.0 ↓20.9	31.0 ↓56.9	50.5 ↓37.4	17.6 ↓70.3	63.9 ↓24.0
WorldVLA	79.1	0.3 ↓78.8	30.2 ↓48.9	44.2 ↓34.9	29.4 ↓49.7	14.5 ↓64.6	12.2 ↓66.9	39.4 ↓39.7
UniVLA	95.2	4.3 ↓90.9	50.3 ↓44.9	71.8 ↓23.4	59.1 ↓36.1	80.0 ↓15.2	25.3 ↓69.9	34.3 ↓60.9
RIPT-VLA	97.5	58.3 ↓39.2	36.7 ↓60.8	80.1 ↓17.4	87.9 ↓9.6	90.4 ↓7.1	73.8 ↓23.7	76.5 ↓21.0

LIBERO-Plus: In-depth Robustness Analysis of Vision-Language-Action Models

Limitations of Current Approaches

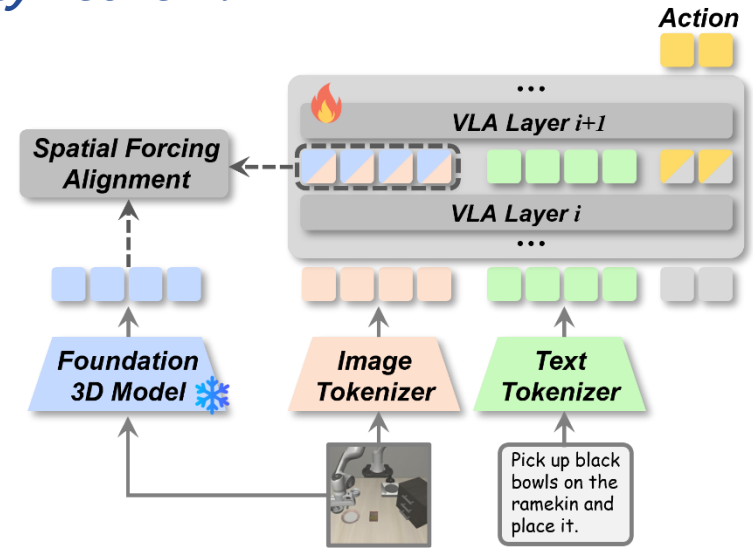
Strong semantic or temporal priors still leave *geometry* implicit.

- **VLMs and WAMs are trained fundamentally on 2D;**
  action decoder should disentangle 3D cues (depth, scale, occlusion) alone
- **2D VLMs struggles across environment changes, such as camera viewpoint change**
- ***3D Geometry can enhance the generalizability of the policy network.***



(A) SpatialVLA

Represent current observation as 3d



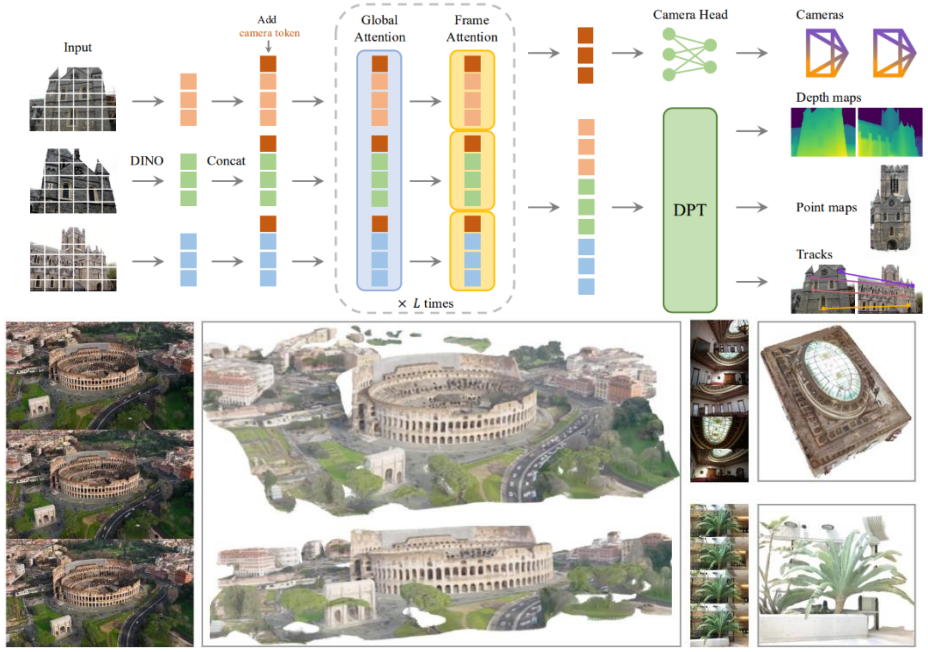
(B) Spatial Forcing

Distill the knowledge of *Geometric Foundation Model*

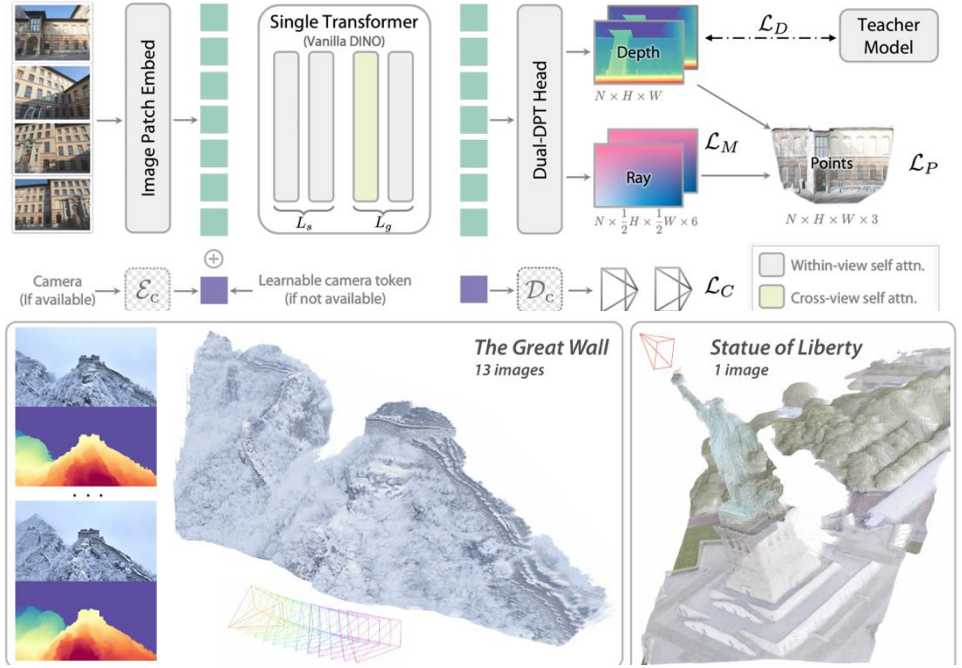
Motivation: Repurposing Geoemtric Foundation Models

Geometric Foundation Models

- **Trained from large-scale multi-view data**
- **predicts accurate 3D-aware representations (Depth / Camera / Track) of scenes**
- **Their internal features shows high cross-view correspondence and geometry awareness for 3D consistency**



(a) VGGT

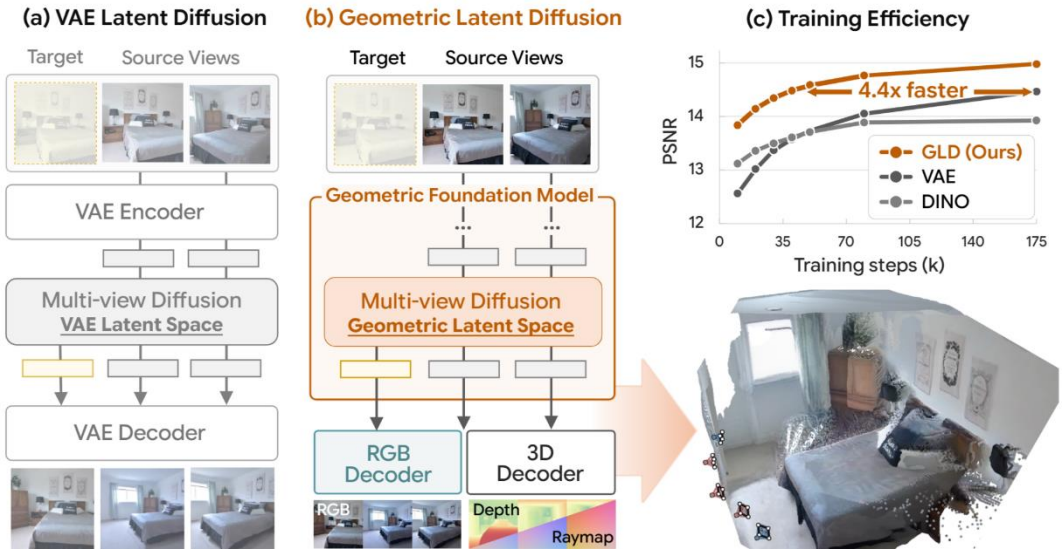


(b) Depth Anything 3 (DA3)

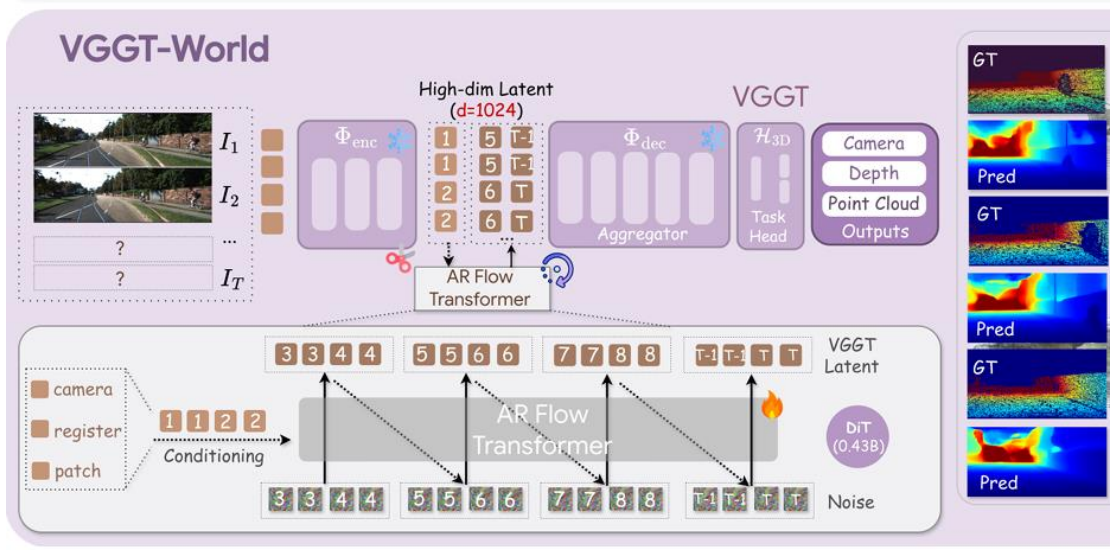
Motivation: Repurposing Geometric Foundation Models

Geometric Foundation Models

- Trained from large-scale multi-view data
- predicts accurate 3D-aware representations (Depth / Camera / Track) of scenes
- Their internal features shows high cross-view correspondence and geometry awareness for 3D consistency
- Recent works show that GFMs itself can be repurpose for multi-view diffusion / world modeling



(a) GLD



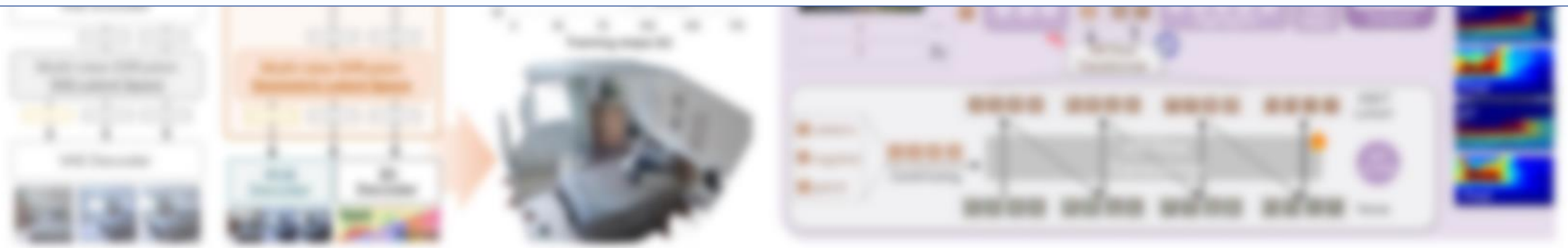
(b) VGGT-World

Motivation: Repurposing Geometric Foundation Models

Geometric Foundation Models

- Trained from large-scale multi-view data
- predicts accurate 3D-aware representations (Depth / Camera / Track) of scenes
- Their internal features shows **high cross-view correspondence** and **geometry awareness** for 3D consistency

Rather than using GFMs only as external feature extractors for policy learning,
Can GFMs itself be repurposed as a robot manipulation policy?

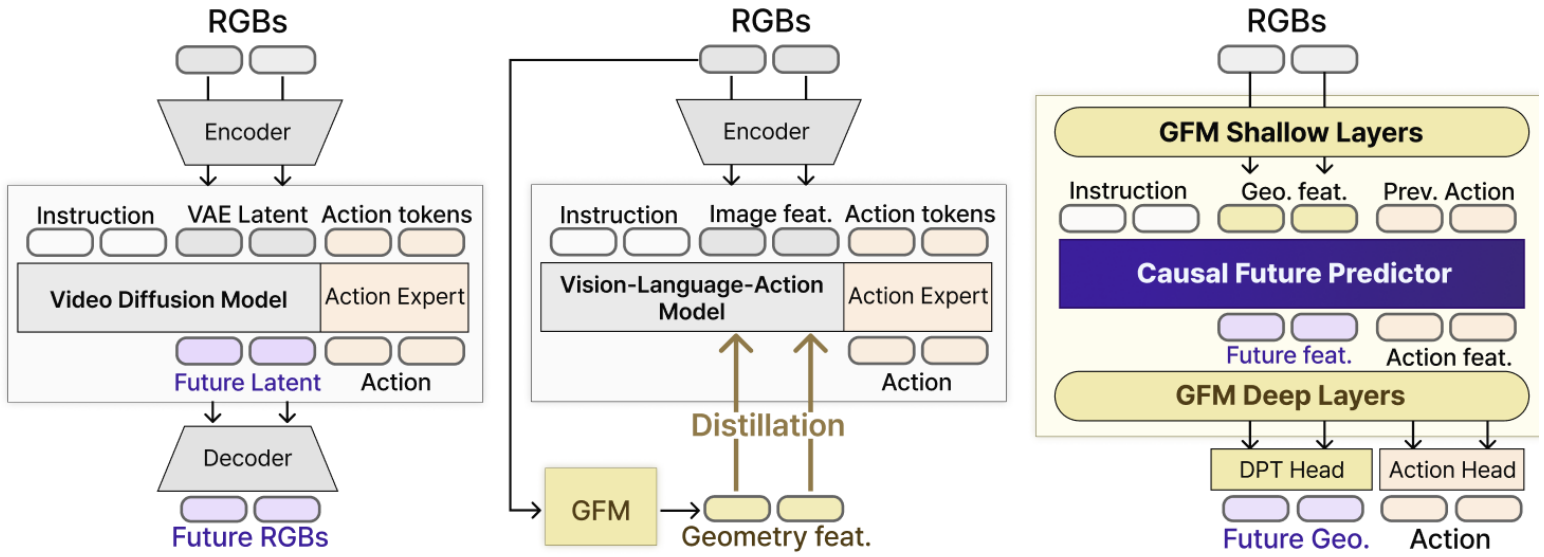


(a) GLD (b) VGGT-World

Geometric Action Model

Geometric Action Model (GAM)

- **Repurpose a pretrained Geometric Foundation Model as a shared backbone for (1) observation encoding, (2) future geometry prediction, and (3) action decoding.**



VDMs → WAMs

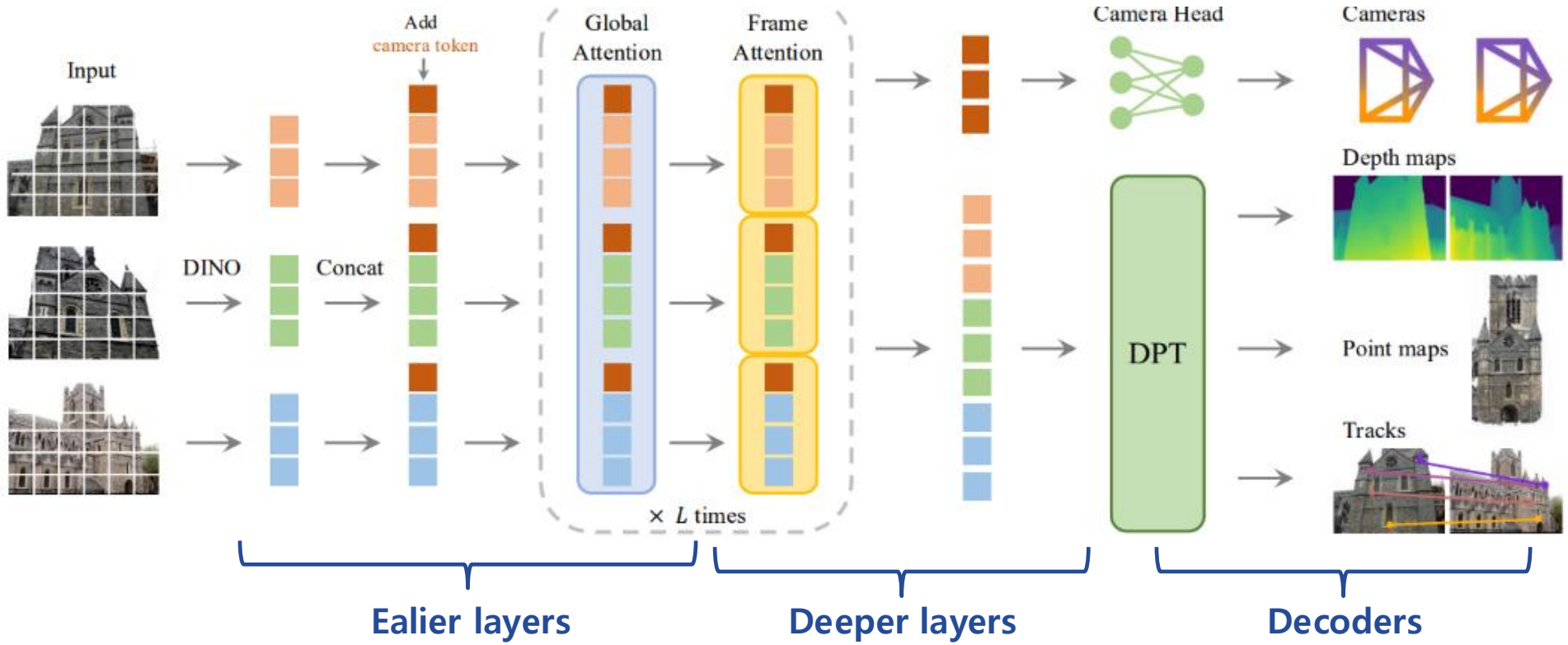
VLMs → VLAs

GFMs → GAMs

Prelimiaries: Geometric Foundation Models

Geometric Foundation Models

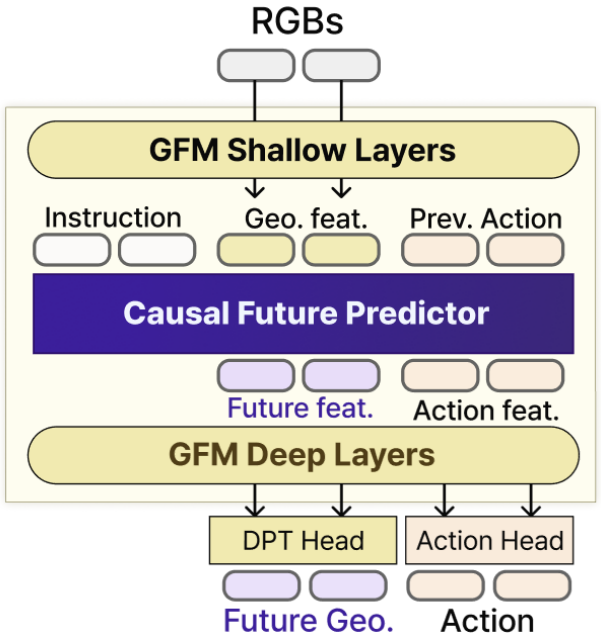
- **Transformer backbone consist of alternationg attention layers; Global attention & Frame attention**
- **DPT head predicts depth/point/track for multi-view input**
- **Camera tokens and shallow camera token predicts the camera pose of input views**



GAM Pipeline

Geometric Action Models

- **Input: multi-view RGB, proprioception, previous actions, task instruction**
- **Output: an 8-step action chunk and optional future geometry**
- **A causal future predictor forecasts latent geometric tokens**

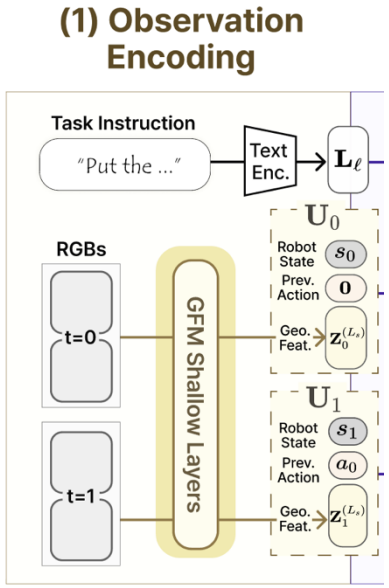


(c) GAM

GAM Pipeline

Observation Encoder

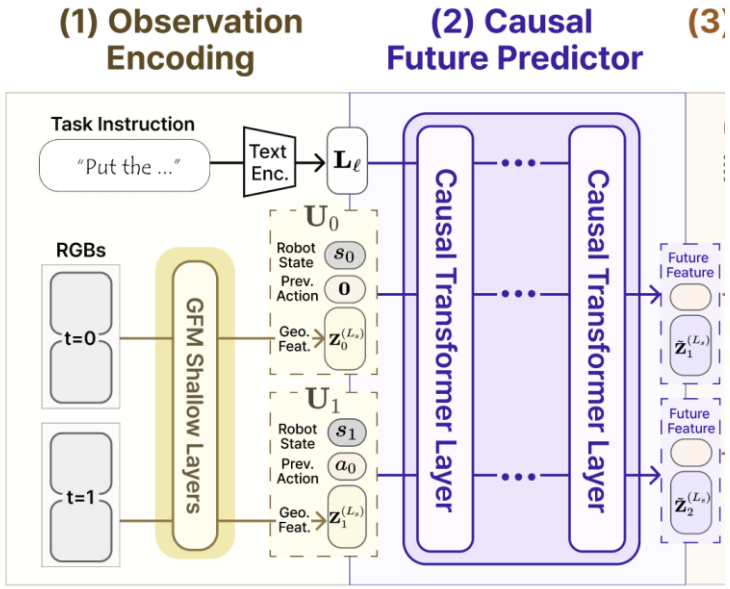
- Use shallow GFM layers as an observation encoder
- 2 view (external + wrist view) RGB input (+ optional RGB history)
- Split layer: 12(out of 40), before the earliest DPT decoding layer
- Encode task instruction, previous action, proprioception, too.

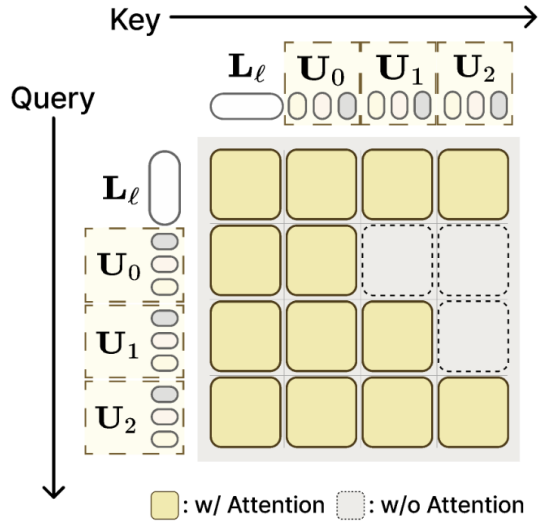


GAM Pipeline

Causal Future Predictor

- **Input: feature of current/past observations, language, proprioception, and previous action**
- **Output: future geometric latent tokens and an action latent token**
- **When the past observations are given, apply block causal attention mask to prevent future leakage**



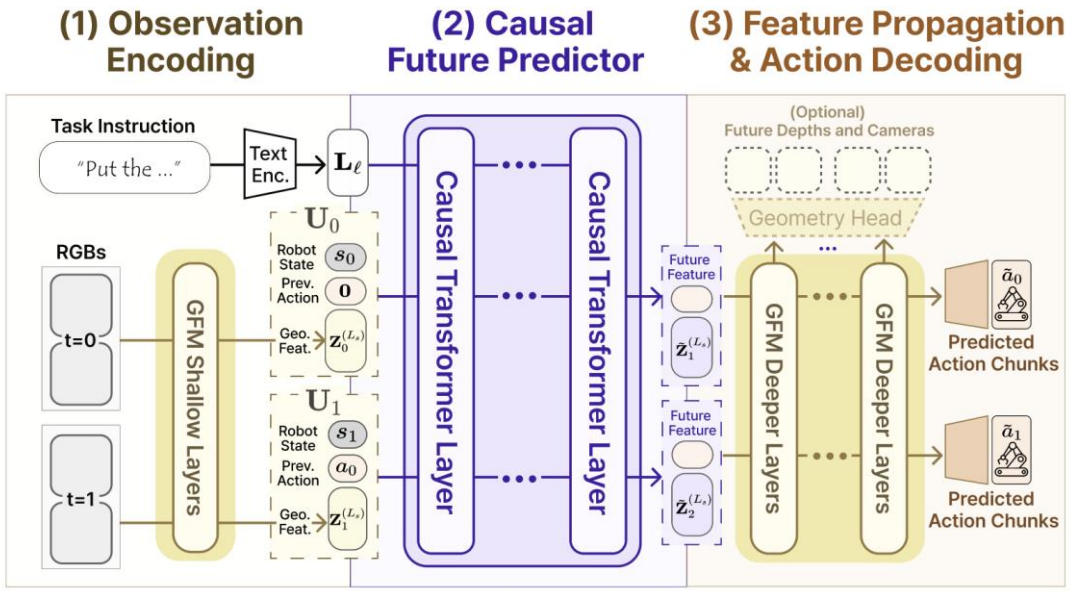


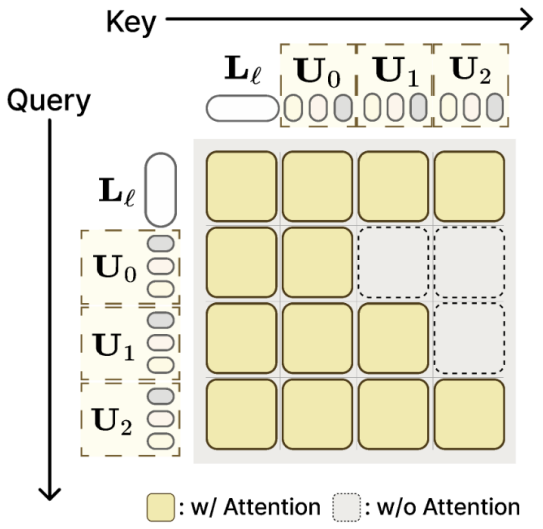
(b) Block-causal Attention Mask

GAM Pipeline

Feature propagation and Action decoding

- **Predicted future tokens are passed through the remaining GFM layers**
- **A lightweight action head regresses the executable action chunk**
- **The original depth head decodes action-aligned future depth maps**



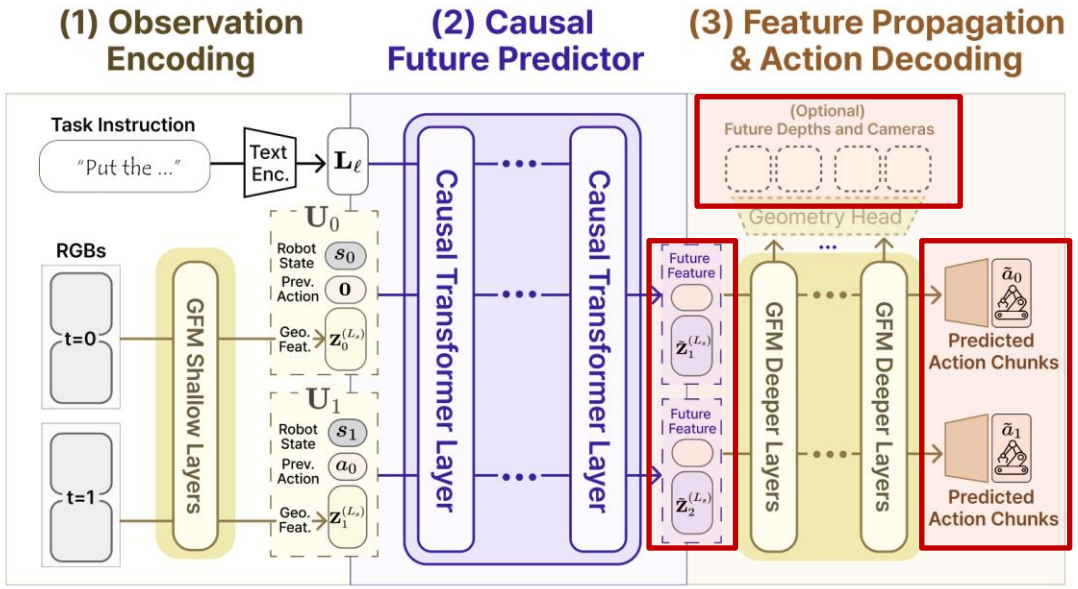


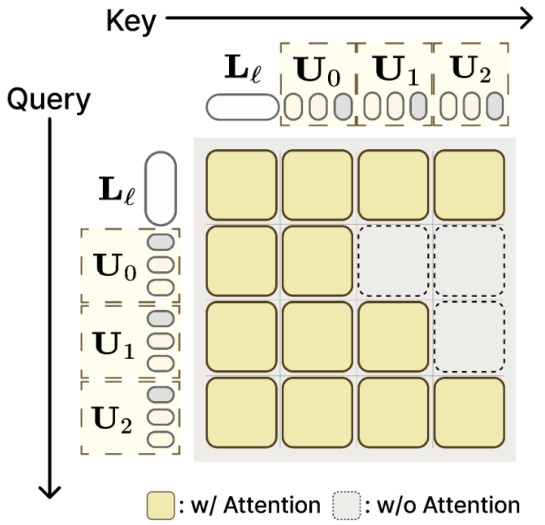
(b) Block-causal Attention Mask

GAM Pipeline

Loss function

- **Future prediction losses**
  - **Future Feature loss (L1)**
  - **Future Depth loss; Depth heads are only used for supervision; can be skipped during inference**
- **Action chunk prediction loss**
  - **7D regression loss**



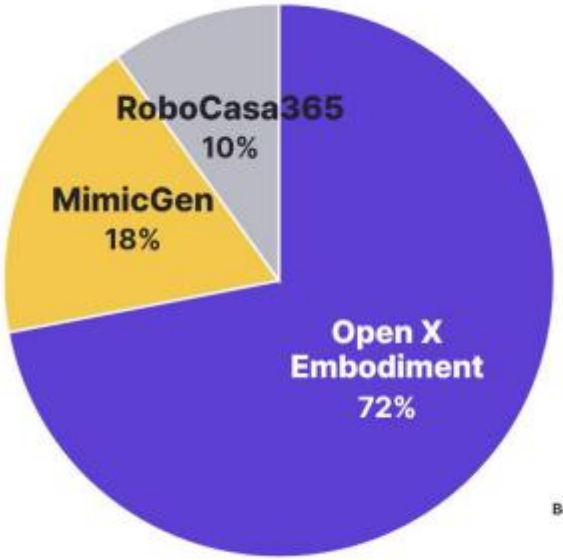


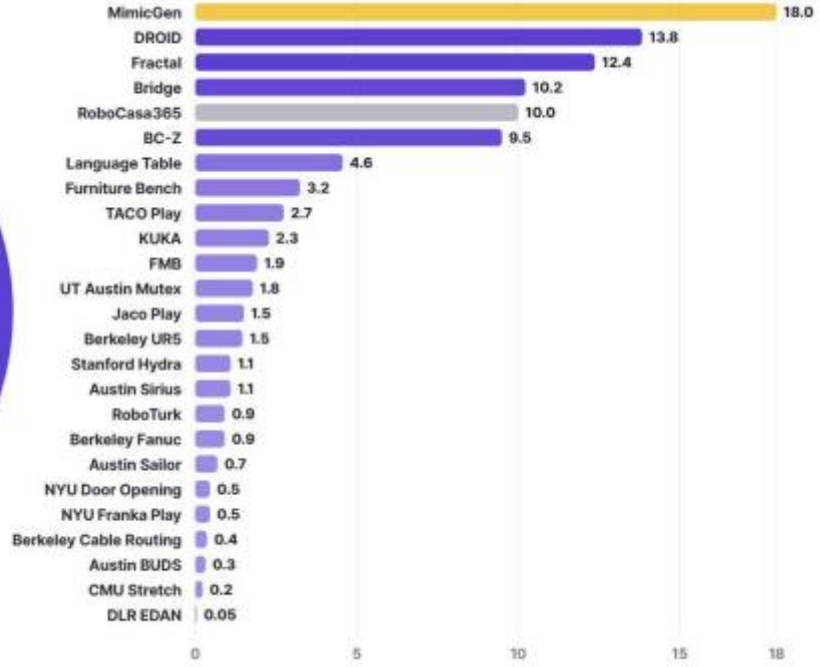
(b) Block-causal Attention Mask

Training Details

Training Details

- **Initialization: DA3-Giant finetuned for 4D reconstruction on Track4World**
- **Pretraining: 784k single-arm robot trajectories; Open-X Embodiment, MimicGen, Robocasa365**
- **RGB History: H = 4 for pretraining, H =1 for post-training**





Experimental Setup

Evaluation

- **LIBERO: standard manipulation benchmark**
- **LIBERO-Plus: zero-shot perturbations in camera, robot, language, lighting, background, noise, and layout**
- **Real robot: four tasks, wrist-mounted and third-person cameras**
- **Robocasa-kitchen: Harder & Longer manipulation tasks**

Baselines: VLAs, world-action models, and geometry-aware VLAs

Simulation Benchmarks

- **LIBERO:**
  - **competitive with baseline; while very satureated benchmeark**
- **LIBERO-Plus:**
  - **SOTA-level generalization capability with only 1.4B parameter.**
  - **Significant improvement on camera perturbation (+9.7%p)**
- **Robocasa-kitchen**
  - **Achieved better performance compared to other VLA/WAM baselines**

(a) LIBERO & LIBERO-Plus

Method	Size	Orig.	Plus	Cam.	Robot	Lang.	Light	BG	Noise	Layout
VLAs										
$\pi_{0.5}$ [5]	3.3B	96.9	84.6 (↓12.3)	72.0	76.6	86.5	96.1	95.2	86.7	86.0
OpenVLA-OFT [3]	7B	97.1	69.6 (↓27.5)	56.4	31.9	79.5	88.7	93.3	75.8	74.3
RIPT-VLA [31]	7B	97.5	68.4 (↓29.1)	55.2	31.2	77.6	88.4	91.6	73.5	74.2
π_0 [39]	3.3B	91.3	69.3 (↓22.0)	61.0	40.8	63.7	89.3	84.1	80.1	75.9
π_0-FAST [6]	3.3B	85.5	61.6 (↓23.9)	65.1	21.6	61.0	73.2	73.3	74.4	68.8
UniVLA [30]	8.5B	95.2	42.9 (↓52.3)	1.8	46.2	69.5	69.0	81.0	21.2	31.9
NORA [23]	3B	87.9	39.0 (↓48.9)	2.2	37.0	65.1	45.7	58.6	12.8	62.1
OpenVLA [2]	7B	76.5	15.6 (↓60.9)	0.8	3.5	23.0	8.1	34.8	15.2	28.5
WAMs										
Cosmos-Policy [7]	2B	98.5	82.4 (↓16.1)	73.4	63.3	89.3	98.9	83.5	89.3	84.0
Fast-WAM [50]	6B	97.6	50.0 (↓47.5)	16.4	44.5	68.9	78.2	53.7	37.7	60.7
WorldVLA [32]	7B	79.1	25.0 (↓54.1)	0.1	27.9	41.6	43.7	17.1	11.0	38.0
Geometry-aware VLAs										
$\pi_{0.5}$ + Spatial Forcing [16]	3.3B	94.0	25.7 (↓58.3)	0.1	0.3	26.8	66.0	45.9	0.1	59.8
$\pi_{0.5}$ + ROCKET [17]	3.3B	95.3	47.5 (↓46.6)	30.9	75.6	29.3	69.2	47.0	25.4	62.0
GAM										
Ours	1.4B	97.6	85.5 (↓12.1)	83.1	70.0	84.8	97.2	94.3	95.3	79.1

(b) Robocasa-Kitchen

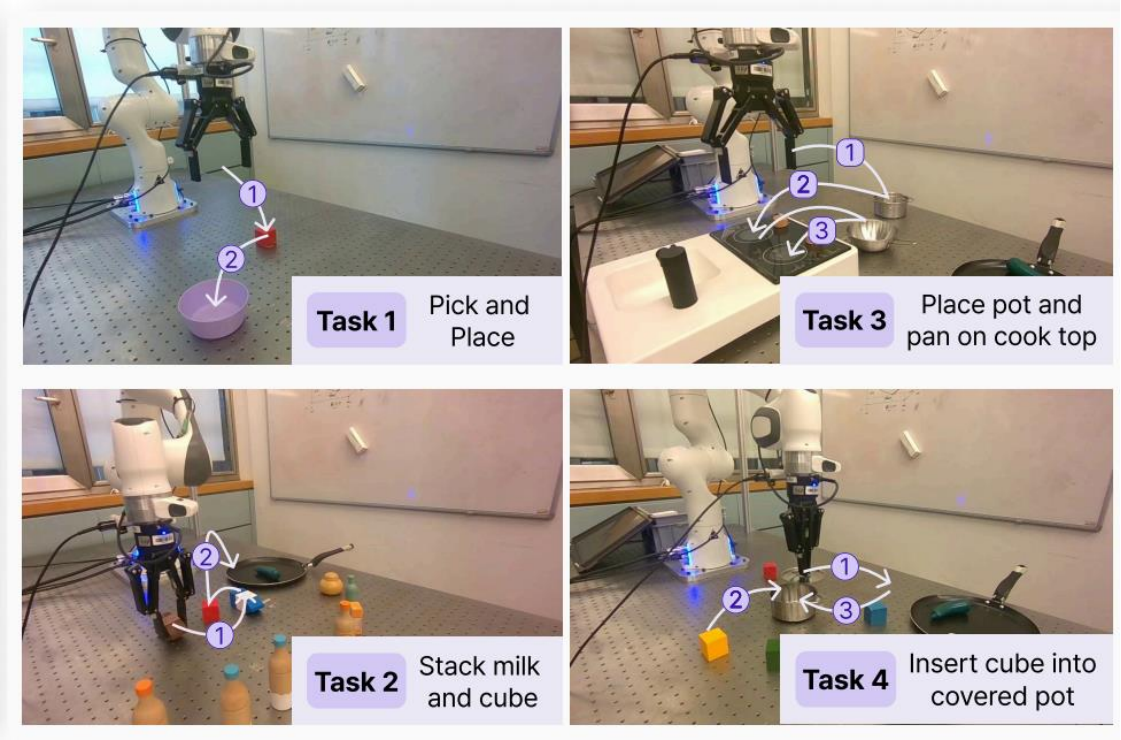
Method	Avg. SR (%)
GROOT-N1	49.
+ DreamGen	57.
+ DUST	58.
UWM	60.
π_0	62.

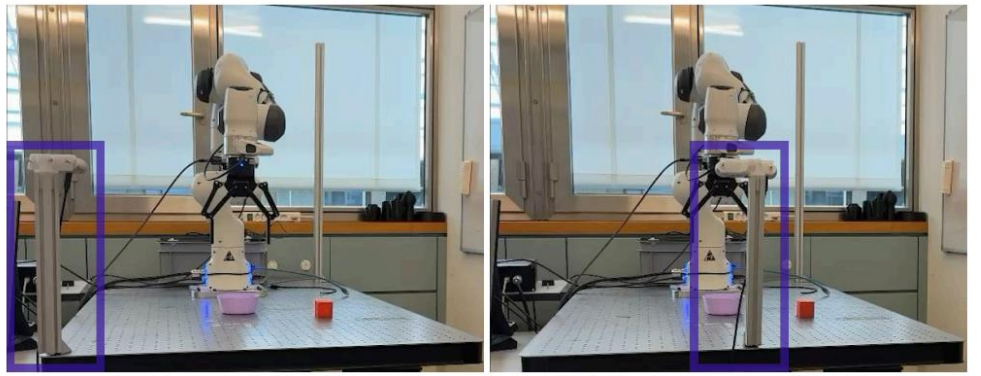
Method	Avg. SR (%)
GROOT-N1.5	64.1
+ HAMLET	66.4
Video Policy	66.0
FLARE	66.4
Cosmos Policy	67.1
GAM (Ours)	**69.4**

Real-world Evaluation

Real-world evaluation

- For real-world evaluation, we defined 4 tasks and collected 200 demos for each task, then post-trained baseline and our models.
- To examine robustness to camera viewpoint change, we evaluated under ID/OOD setting
- Baselines: Pi0.5, Spatial Forcing



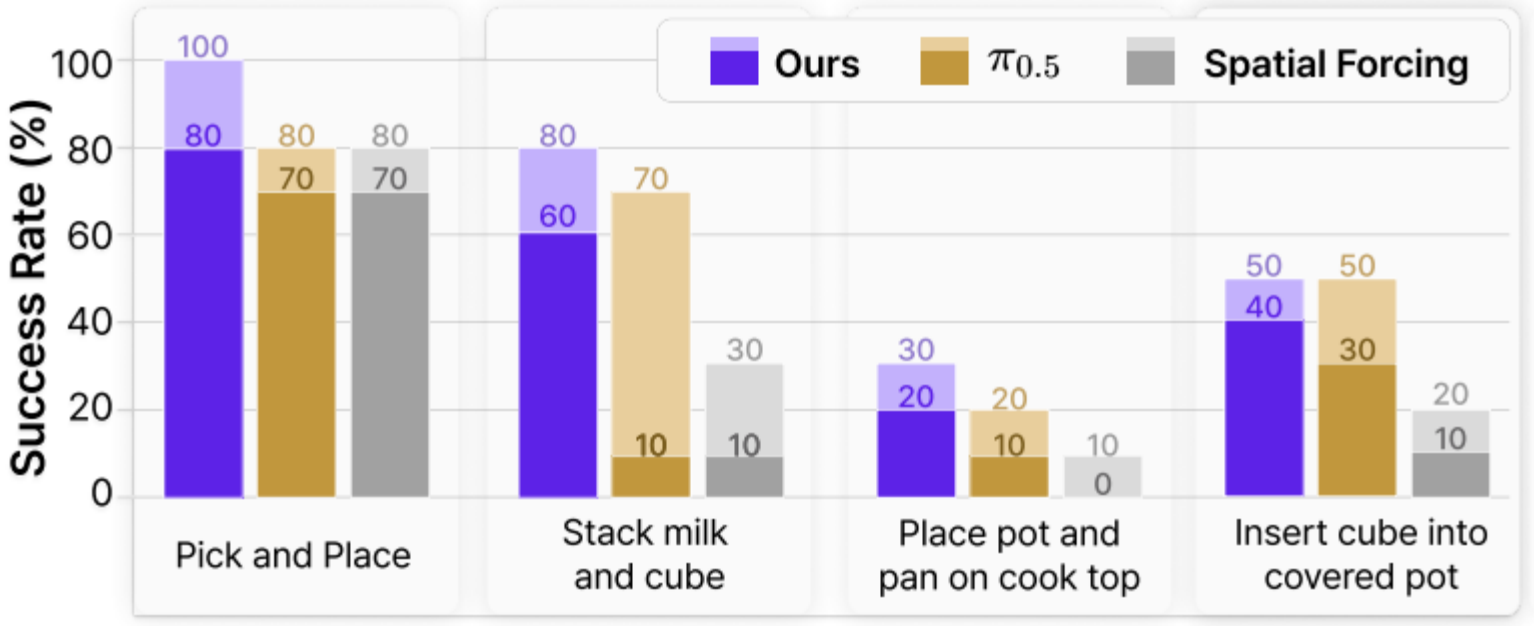


(a) In-distribution (b) Out-of-distribution

Real-world Evaluation

Real-world evaluation

- **For real-world evaluation, we defined 4 tasks and collected 200 demos for each task, then post-trained baseline and our models.**
- **To examine robustness to camera viewpoint change, we evaluated under ID/OOD setting**
- **Baselines: Pi0.5, Spatial Forcing**
- **GAM achieves higher ID/OOD success rate compared to baselines**

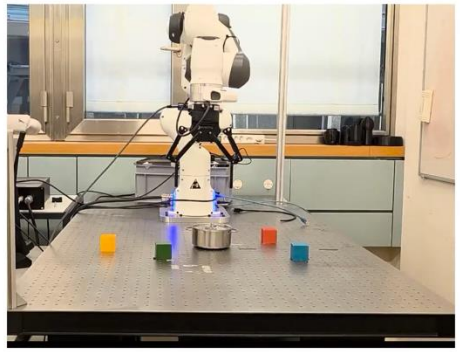


Real-world Evaluation

4. Insert cube into covered pot

x 10

Trial 1



Trial 2

GAM

Pi 0.5

Spatial Forcing

Further Results and Analysus

Inference Latency

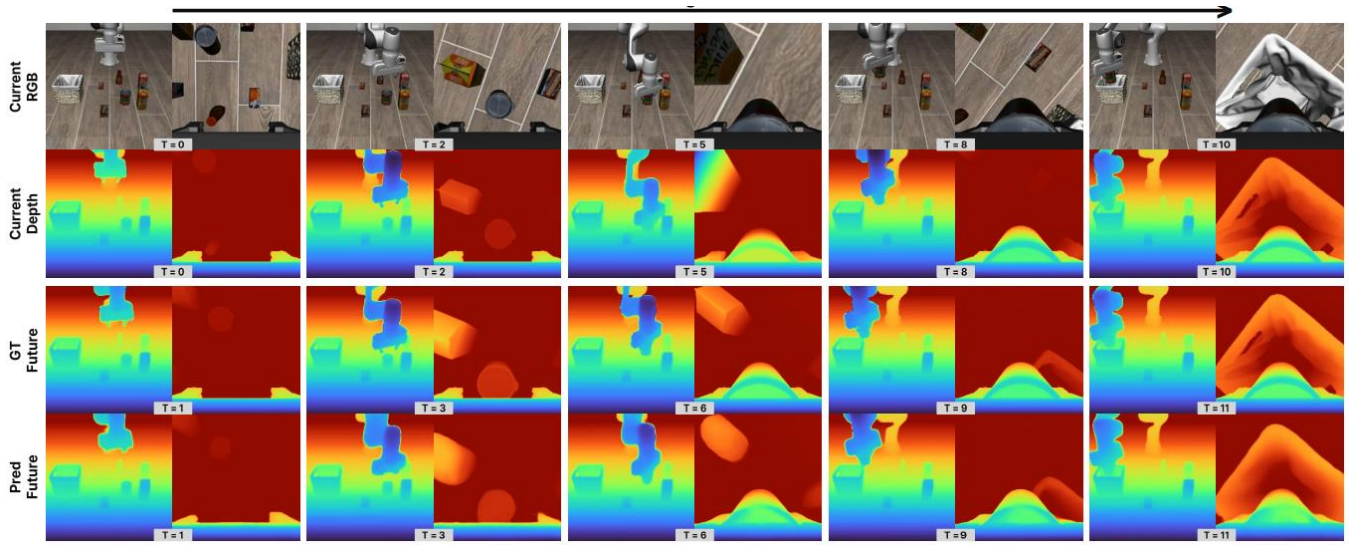
- **GAM achieves ~4 – 50x faster inference speed comapared to VLM / Video Diffsion based methods**

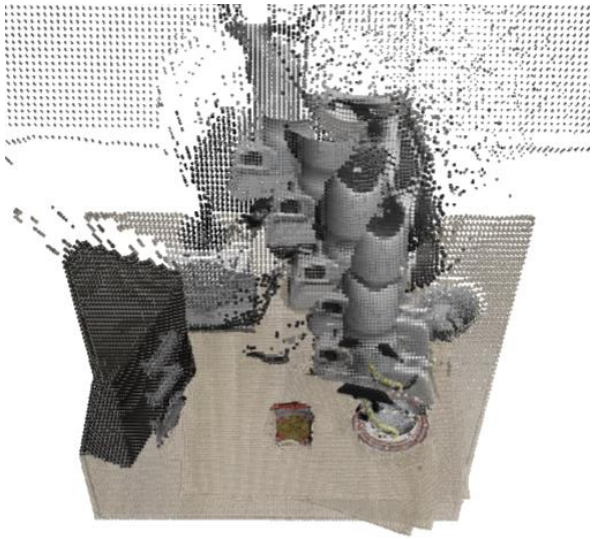
Table 8: **Model-only inference latency on a single GH200 GPU.**

Policy	Official PyTorch	bf16 precision	Torch Compile	CUDA Graphs	Model-only latency
GAM (Ours)	✓	✓	✓	✗	17.5 ms
GAM (Ours)	✓	✓	✓	✓	**6.9 ms**
pi0.5	✓	✓	✓	✗	29.2 ms
OpenVLA-OFT	✓	✓	✓	✗	70.1 ms
Cosmos Policy	✓	✓	✓	✗	382.4 ms

Future Depth & 3D Reconstruction

- **Since future predictor of GAM predicts native future feature of GFMs, we can decode future depth and 3D scenes**





Discussion and Conclusion

Takeaway

- **GFMs can be more than feature extractors: they can become the policy substrate for predicting both future scene geometry and robot actions.**
- **GAM is efficient, fast, and strong policy network with its single-pass inference and unified modeling with geometric backbone, especially strong on geometric perturbation; camera viewpoint change**

Future Directions & Discussion

- **How geometric prior of GFM influences the robustness of the policy modeling?**
- **Compared to VLAs and WAMs, high-level reasoning capability for more complex task can be limited.**

Outline

- About me
- Introduction
- From Spatial AI to Physical AI: Learning to Represent, Reason, and Act in 3D
  - C3G: Learning Compact 3D Representations with 2K Gaussians
    - CVPR 2026 # Compact 3D Gaussian Splatting # 3D Reconstruction # 3D Scene Understanding
  - Imagine3D-LLM: Teaching MLLMs to Imagine 3D Scenes Before Answering
    - NeurIPS 2026 (under review) # Compact 3D tokens # Mental Reconstruction # Spatial Reasoning
  - Geometric Action Model for Visuomotor Control
    - Arxiv 2026 (under review) # WAM # VLA # Manipulation # Geometric Foudnation Model # World Model
- **Conclusion**
- Future Works

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- **Current robotics area is accepting learning-based approaches**
  - Easier for us computer vision people to start tackling, without having to know control or mechanical engineering backgrounds
  - However, from industries and academia, those who are familiar with both the vision + language + control are being prioritized

Outline

- About me
- Introduction
- Spatial AI: From 2D vision to 3D vision
  - Emergent Outlier View Rejection in Visual Geometry Grounded Transformers
    - CVPR 2026 # VGGT # Image Retrieval # 3D Reconstruction
  - C3G: Learning Compact 3D Representations with 2K Gaussians
    - CVPR 2026 # Compact 3D Gaussian Splatting # 3D Reconstruction # 3D Scene Understanding
  - LitePT: Lighter Yet Stronger Point Transformer
    - CVPR 2026 Highlight # Point Cloud Backbone # Efficiency
- Conclusion
- **Future Works**

Future Works



Thank you!

https://sunghwanhong.github.io/